From Drought to Perinatal Health: The Role of Agriculture and Air Quality

PRELIMINARY DRAFT

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November 4, 2026

Abstract

Between 2009 and 2012, Mexico experienced one of its most severe droughts in recent decades. Using administrative birth records for more than eight million deliveries, I document that in-utero exposure to severe rainfall deficits reduced birth weight and increased the incidence of low birth weight and preterm delivery by about 2 and 3.5 percent above baseline, respectively. Effects are heterogeneous across geographic climate regions and socioeconomic conditions, with the strongest effects concentrated in poor rural municipalities of humid regions, though urban areas are also affected. Moreover, I document negative effects on maternal health, with about a 7 percent increase in admissions for pregnancy complications, and only in arid and semi-arid municipalities. Informed by this heterogeneity, I analyze agricultural outcomes and air quality as potential mechanisms. I find that it would require sizeable damages to agricultural production, but only a small increase in $PM_{2.5}$ concentration, to explain the whole effect of the drought event. The results suggest a role for prenatal and obstetric care, and for air quality management, in drought policy.

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1 Introduction

During 2009–2012 Mexico experienced one of its most severe drought episodes in decades. The episode reached its peak in 2011 and covered close to seventy percent of the territory, generating economic losses estimated up to around ten percent of GDP (Arceo-Gómez et al. 2020). Although shocks of this magnitude are of general health policy concern, the perinatal period is specially sensible as even transitory shocks can have long-lasting consequences for cognition, health, and adult earnings (Almond and Currie 2011; Cunha and Heckman 2007; Heckman and Mosso 2014; Adhvaryu et al. 2023).

The literature on environmental hazard and perinatal health has focused mainly on temperature (Salvador et al. 2023). Indeed, it has been documented that in-utero exposure to heat reduces birth weight and increases the incidence of preterm delivery (Deschênes et al. 2009; Le and Nguyen 2021; Andriano 2023; Hong 2025). In contrast, severe drought events have received less attention, even though their effects and mechanisms differ from those of temperature alone (Meherali et al. 2024). Although some studies on the relationship between drought and perinatal health exist, these often focus on specific climatic regions or socioeconomic groups (Rocha and Soares 2015; Lin et al. 2021), overshadowing differences across populations within a same jurisdiction. Moreover, the mechanisms from drought to perinatal health are far less understood, with agricultural pathways often hypothesized but rarely formally tested (Maccini and Yang 2009; Rocha and Soares 2015; Lin et al. 2021; Abiona and Ajefu 2023; Rocha et al. 2026). This paper investigates whether and to what extent in-utero exposure to the 2009–2012 meteorological drought in Mexico affected perinatal health, analyzes the spatial and socioeconomic heterogeneity of the health burden, and explores agricultural outcomes and air quality as potential mechanisms whose relative importance may differ across regions.

To conduct this analysis, I build a unique combination of administrative datasets that cover the whole Mexican territory over the study period and let me analyze meteorological droughts from different independent sources. The main dataset comes from administra-

tive birth certificates, which provide near-census coverage of live-born deliveries across the country. I complement these data with hospital discharge records for mothers, agricultural production data, and satellite-based air pollution measures.

Using a recentered rainfall deficit measure (Borusyak and Hull 2023) to capture meteorological drought during gestation, I estimate that at the national level one log point of shortfall reduces birth weight by 7.7 grams and raises the incidence of low birth weight and preterm delivery by 0.18 and 0.30 percentage points, respectively. To put magnitudes in context, a severe deficit of 50% less rainfall than expected would reduce birth weight by 5.3 grams and increased the incidence of low birth weight and preterm delivery by 125 and 208 cases per 100,000 births, or about 2.2% and 3.4% above baseline during the period of analysis.[1]

I document that the effects of rainfall deficits are heterogeneous across geographic regions and socioeconomic groups. Particularly, rural municipalities in humid regions, the poorest in the sample and the most dependent on rainfed agriculture, show a birth weight marginal response close to three times the national estimate. In turn, the incidence preterm delivery, responds strongly in urban municipalities. Surprisingly, I find no detectable effect on birth outcomes in arid areas. Yet maternal hospitalizations increase by about 7

Nonetheless, marginal effect (sensitivity) to rainfall is only one dimension to understand the health burden of the episode. Following the IPCC's SREX risk framework (Cardona et al. 2012), I identify the burden as the product of location-specific sensitivity, the magnitude of the event, and the size of the population exposed. Specifically, I estimate that the burden of 2011, the most severe year, concentrated in the north-east of the country, covering both dry and non dry municipalities in the highlands of the Sierra Madre, where a high share of indigenous populations reside. Although the national average loss is 2.4 grams per birth, it is around 13 to 15 grams in the most affected municipalities of Chihuahua and Durango. For preterm delivery, the median municipality records a burden of 0.7 percent of its expected

[1]The India Meteorological Department classifies a seasonal rainfall deficit of more than 50% of the normal as severe drought (`https://mausam.imd.gov.in/imd_latest/monsoonfaq.pdf`)

cases, and up to 6.8 percent in the most affected regions. What is more, the same level of burden can arise for different reasons. For instance, in Monterrey a loss of about 5 grams per birth comes mostly from a severe deficit, while in the rural highlands of the State of Mexico and Veracruz a loss of similar size arises mostly from local sensitivity to rainfall deficits.

Informed by this heterogeneity, I explore agricultural outcomes and air quality deterioration as potential mechanisms. Under a single-channel assumption, an instrumental variables design allows me to place an upper bound on each mechanism. Results suggest that a drought-induced rise of 6% in $PM_{2.5}$ concentrations would produce the same birth-weight loss as a rise of 140% in crop losses, relative to their baseline. Similar results hold for low birth weight and preterm delivery.

In other words, a large change in agricultural productivity would be required to produce the same effect as a small change in air quality. This is consistent with evidence that agricultural households in middle-income countries are able to smooth part of the shock through credit, informal insurance and transfers, even if imperfectly (Hou 2010; Edwards et al. 2024). In contrast, evidence suggests that even low-concentration but sustained in-utero exposure to air pollution can affect birth outcomes (Currie et al. 2009; Arceo et al. 2016; Shi et al. 2024).

These results have implications for how drought policy is designed. Since both the effects and the burden of the episode differ across regions, a uniform national response may not address the needs of the most affected populations. Particularly, the rise in maternal hospitalizations in dry regions suggests that access to prenatal and obstetric care could be important in these areas. Moreover, although drought preparedness in Mexico was built around agricultural support and emergency relief, measures addressing air quality during dry spells, such as dust suppression, wildfire prevention and particulate monitoring, could complement this approach.

2 Literature review

Research on the implications of climate on perinatal health has expanded rapidly as climate datasets become more readily accessible. Indeed, evidence suggest that exposure during pregnancy can reduce birth weight, raise the risk of preterm delivery, and increase neonatal mortality, with evidence from both high-income (Deschênes et al. 2009) and developing countries (Andalón et al. 2016; Le and Nguyen 2021; Andriano 2023). The identified channels are both physiological and behavioral or contextual (Ebi et al. 2021).[2]

In Mexico, the most developed line of this research also concerns temperature. Cohen and Dechezleprêtre (2022) document that extreme heat increases mortality across the country, and that the rollout of Seguro Popular attenuated this effect, suggesting that public health provision can moderate the cost of weather stress. Closer to the outcomes studied here, Hong (2025) draw on the same administrative birth certificate records and find that in-utero exposure to wet-bulb temperatures reduces birth weight and shortens gestation.

Compared to the more developed body of research on heat, studies on the effects of drought on perinatal health remain limited and less cohesive (Ebi and Bowen 2016; Salvador et al. 2023). Indeed, recent reviews consistently identify drought as an understudied determinant of maternal and infant health, particularly in low- and middle-income settings where adaptive capacity is often insufficient (Meherali et al. 2024). In addition, I am not aware of any meta-analysis of drought and perinatal health, a form of synthesis that is standard for other environmental exposures.

A large mass of the evidence of the relationship between drought and perinatal health come from correlational and qualitative studies (United Nations Development Programme 2006; World Health Organization 2012; Algur et al. 2021; Desai and Zhang 2021; Ponnusamy 2022; Diamond-Smith et al. 2023). A small body of causal evidence from low- and middle-income countries documents that rainfall deficits negatively affect birth weight, gestation

[2]For instance, medical research suggests that pregnancy reduces thermoregulatory capacity, so elevated temperatures can limit placental perfusion, trigger hormonal responses that raise the risk of early labor, and disrupt immune function (Conti et al. 2020; Ha 2022)

time, and child mortality, with evidence spanning Brazil, China, Sierra Leone, among others (Maccini and Yang 2009; Rocha and Soares 2015; Lin et al. 2021; Abiona and Ajefu 2023). Although effect sizes vary depending on the identification strategy, exposure measure, and population studied, the direction of the association is consistent, with rainfall deficits negatively affecting perinatal health.

To my knowledge, no research for Mexico has studied the effects of drought on perinatal health at the national level. Existing studies on rainfall shocks and early life with data for Mexico focus on outcomes measured well after birth, including child height, cognition, schooling, and adolescent employment, with evidence mainly from rural or low-income populations (Skoufias and Vinha 2012; Aguilar and Vicarelli 2022; Adhvaryu et al. 2023). Close to this paper, Arceo-Gómez et al. (2020) studies the same unexpected drought event in Mexico (2009-2012) and reports lower rural earnings, higher poverty levels, and reduced female labor force participation and school attendance. As in the international literature, the mechanism most frequently mentioned across these papers is agricultural disruption and the household income loss it produces.

I contribute to the literature in several ways. First, from a methodological point of view, I construct a recentered rainfall deficit measure during gestation in the sense of Borusyak and Hull (2023) as the main exposure variable. Most causal studies identify the effect of drought from within-location variation in rainfall, treating the shock as good as randomly assigned conditional on location, season, and time fixed effects. However, omitted variable bias can arise from nonrandom exposure to the exogenous shock. In other words, even when rainfall is as good as randomly assigned, municipalities can differ in unobserved determinants of birth outcomes in ways that are correlated with their exposure to it, which the fixed effects cannot absorb (e.g., self-selection into certain locations). Recentering removes this source of bias, and pins down the effect from comparing municipalities above and below their own expected counterfactual rainfall.

Specifically, the identifying assumption is that, conditional on geographical, time, and

demographic characteristics, the rainfall deviation is uncorrelated with unobserved determinants of birth outcomes. Intuitively, this shifts the comparison from municipalities that received more or less rainfall to municipalities above and below their own expected counterfactual.

Second, I show that the response to drought is spatially heterogeneous, varying across climate regions and urbanization settings. Most studies focus on a single region, whether the semiarid Northeast of Brazil (Rocha and Soares 2015), one Chinese province (Lin et al. 2021), or rural Sierra Leone (Abiona and Ajefu 2023). Yet in a country as geographically diverse as Mexico, there is little reason to expect the shock to have homogeneous effects. Furthermore, following the disaster risk framework of the Special Report on Managing the Risks of Extreme Events and Disasters (Cardona et al. 2012), I show that this variation in vulnerability does not by itself locate the burden of the episode. Combining hazard, exposure and vulnerability shows where the burden concentrated and which municipalities were most affected.

Third, I causally analyze the relative importance of agricultural disruption and air quality as mediators of the effect of drought on perinatal health. In this literature, agricultural disruption is widely accepted as one of the main mechanisms through which a weather shock affects health in a middle-income country, although the chain from drought to agricultural production to health rests mostly on correlational evidence. No causal estimate unequivocally attributes drought-induced agricultural variation to perinatal health.[3] Competing pathways are explored still less. In particular, the environmental literature establishes that drought can raise particulate concentrations through wildfire, dust mobilization from dry soils, and reduced wet deposition (Ebi et al. 2021; Salvador et al. 2023). What is more, a separate body of work documents the effects of particulate exposure on birth outcomes (Currie et al. 2009). How much of the drought effect runs through each channel is therefore unknown.

In this paper, following the causal mediation model of Dippel et al. (2022), based on in-

[3]Rocha and Soares (2015) examines water access and sanitation coverage, although that analysis rests on heterogeneity by coverage and does not estimate the effect of the mediator itself.

strumental variables, I analyze these potential mediators in detail. Specifically, I decompose the effect of rainfall deficits on birth outcomes into an indirect effect (IE) running through the mediator and a direct effect (DE) that collects the remaining pathways, including those the available data cannot measure, such as maternal stress and other biological responses.

Finally, I provide evidence that the effects estimated for dry regions may be understated by selection. In particular, I find that over the study period drought raised admissions for maternal complications in these municipalities, with the rise appearing both in abortive outcomes and in threatened preterm labor.

3 Background

Mexico's climate is characterized by large geographic contrasts in rainfall. Average annual precipitation ranges from less than 400 mm in the northern plateau and central highlands to over 2,000 mm in the southern Gulf coast and Chiapas (Rodríguez-Moreno et al. 2021). More than half of the national territory is classified as arid or semi-arid, and in these regions precipitation concentrates between June and September, so that normal conditions already involve prolonged seasonal water deficit (Esparza 2014). Drought itself is not exceptional in Mexico. Historical and dendrochronological records document recurrent episodes over several centuries, often tied to broader climate oscillations such as El Niño and La Niña, which drive drought severity in different parts of the country (Martínez-Austria and Díaz-Jiménez 2018). Seventy-seven percent of the Mexican population lives in zones that hold only 31 percent of available surface water (Rodríguez-Moreno et al. 2021). Figure 1 panel a) maps the distribution of historically dry and non-dry municipalities across the country.

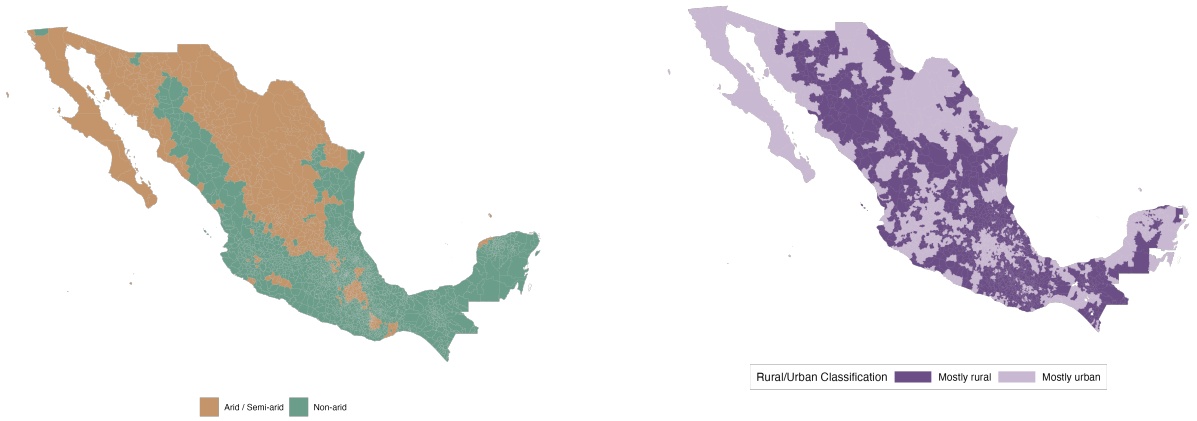
These climatic conditions interact with socioeconomic structure in ways that differ between rural and urban settings (Figure 1 panel b). In rural Mexico, where 22 percent of the population lives, over half of households are poor and agriculture is predominantly rainfed, with only a fifth of cultivated land having access to irrigation (Arceo-Gómez et al. 2020). A

rainfall deficit in these areas can translate directly into lost harvests, lower household income, and food insecurity. Urban and peri-urban populations face a different set of vulnerabilities. Cities in the arid north depend on strained aquifer systems and long-distance water transfers, and drought conditions are associated with increased wildfire activity, soil desiccation, and degraded air quality (Rodríguez-Moreno et al. 2021). The pathways through which a drought affects health are therefore unlikely to be the same across settings, and an analysis that does not distinguish between them risks averaging over distinct processes.

During 2009–2012 Mexico experienced one of its most severe drought episodes in recent history. Dry conditions started in 2009, when a quarter of municipalities already registered severe drought, eased temporarily in 2010, and then escalated durin 2011. By mid-2011, more than 80 percent of municipalities and over 90 percent of the national population lived in areas with some degree of drought, and two-thirds of municipalities experienced severe conditions. Figure 2 shows the geographic spread across the four years. Mean annual precipitation fell to 757 mm, the lowest level recorded since 1956 (Murray-Tortarolo and Jaramillo 2019). The agricultural toll included 2.7 million hectares of maize and beans destroyed and approximately one million cattle dead, and at the household level the shock reduced rural per capita earnings, raised poverty by roughly five percentage points, and shifted women out of employment (Arceo-Gómez et al. 2020). By 2012 the drought had largely receded, with fewer than a fifth of municipalities still under severe conditions.

During the study period, Mexico did not have a comprehensive drought management plan. Indeed, the federal government's main instrument was FONDEN, a disaster relief fund designed for short-term emergency financing after a formal declaration, and the immediate response to the 2011 crisis consisted of ad hoc measures such as water delivery, food distribution, and livestock supplements, with little coordination across levels of government (Ortega-Gaucín et al. 2016). Broader safety-net programs were in place, including conditional cash transfers through Progresa/Oportunidades and the ongoing expansion of public health insurance through Seguro Popular, but none had been designed with drought in mind.

Figure 1: Climate and Urbanization Regions



(a) Climate region (b) Urbanization

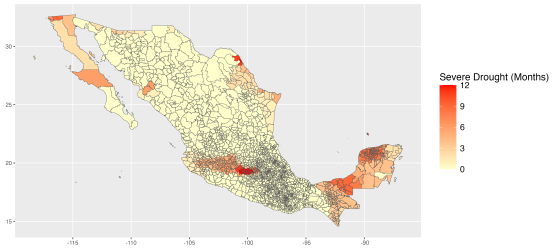
Notes: Panel (a) classifies municipalities as dry (arid and semi-arid) or non-dry (sub-humid and humid) based on the dominant Köppen climate type within municipal boundaries. Source: INEGI climate raster (1902–2011). Panel (b) classifies a municipality as mostly rural when its rural population share exceeds 0.5. Source: INEGI Population Census 2010.

The severity of the episode prompted a shift toward preventive risk management only after the crisis had passed, with Mexico's first national drought program launched in 2013.[4]

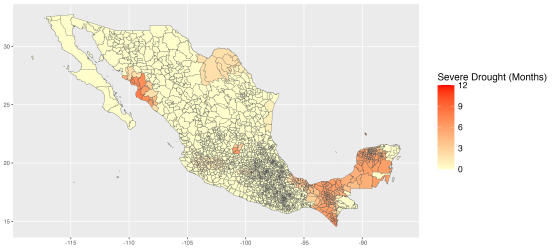
A drought of this magnitude is unlikely to produce uniform consequences in a country where climate, economic structure, and water access vary as widely as they do in Mexico. Arid municipalities with irrigated commercial agriculture face a different shock from humid rural areas dependent on rainfed subsistence farming. In urban centers the relevant exposure may operate through water infrastructure or air quality. The policy tools available during the episode were not designed to address this diversity. The analysis that follows examines how the health effects of the 2009–2012 drought varied across climate regions and urbanization settings, and uses that heterogeneity to point toward the channels at work.

[4]The Programa Nacional Contra la Sequía (PRONACOSE), coordinated by CONAGUA, was announced in January 2013 with the objective of implementing preventive and mitigation actions before drought conditions develop (Ortega-Gaucín et al. 2016).

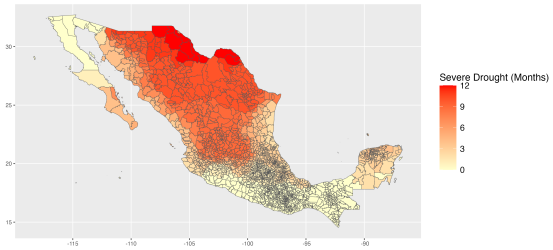
Figure 2: Geographical variation of registered droughts



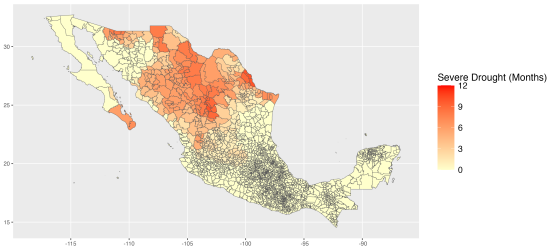
Panel a) 2009



Panel b) 2010



Panel c) 2011



Panel d) 2012

Notes: Darker red indicates 12 months with drought registries in the most severe MSM categories (D2–D4). Yellow indicates no drought registries in those categories.

4 Data and Variables

4.1 Birth outcomes

To analyze birth outcomes, I use individual-level birth certificate records from the Subsistema de Información sobre Nacimientos (SINAC), compiled by the Mexican Ministry of Health (SSA). SINAC records information on each live birth attended at a health facility, including birth weight (in grams), gestational age (in weeks), delivery type, and date of birth. It also reports the municipality of residence of the mother and maternal characteristics such as age, marital status, parity, and schooling. The sample covers 2009–2012 and contains approximately 8 million births across 2,384 municipalities.

Table 1 presents summary statistics by urbanization and climate region. The three outcome variables used in this analysis are birth weight in grams, an indicator for low birth weight (below 2,500 grams), and an indicator for preterm birth (below 37 weeks of gestation). Mean birth weight is 3,163 grams, with a low birth weight rate of 5.6% and a preterm rate of 6.1%. Rural municipalities account for 17% of births but 57% of municipalities (1,355 of 2,384). Mothers in rural areas are younger on average (25.1 vs 25.4 years) and more likely to have primary education or less (39% vs 20%). Preterm rates are lower in rural areas (5.3%) than in urban ones (7.0%), consistent with differences in diagnostic practices and access to early gestational dating.

A known limitation of SINAC is incomplete coverage during its early years of operation. SINAC was launched in 2008 to provide a continuous enumeration of live births through health facility records, but it does not cover births that occur outside medical units (Mier y Terán Rocha and García Guerrero 2019). This undercoverage is concentrated in states with lower access to health services. Table B.4 in the Appendix compares SINAC birth counts against INEGI vital statistics by state for the baseline year, 2009. Most states show coverage rates above 85%, but four southern states fall below 80%: Veracruz (78.7%), Oaxaca (64.9%), Guerrero (64.6%), and Chiapas (53.8%). As a robustness check, I re-estimate the main results

excluding these four states.

Table 1: Summary Statistics

	Rural			Urban		
	All	Dry	Non-dry	All	Dry	Non-dry
N municipalities	1,355	343	1,012	1,029	312	717
N births (2009-2012)	1,420,275	322,813	1,097,462	6,681,902	2,595,131	4,086,771
Panel A: Birth Outcomes						
Birthweight (grams)	3175.9	3182.8	3173.9	3160.6	3188.2	3142.9
Low birthweight (%)	4.67	5.11	4.53	5.81	5.55	5.98
Preterm birth (%)	5.34	6.36	5.04	6.95	6.92	6.96
Gestational weeks	38.99	38.85	39.03	38.76	38.75	38.76
Panel B: Mother Characteristics						
Avg. births per year	355,069	80,703	274,366	1,670,476	648,783	1,021,693
Age	25.10	25.32	25.04	25.40	25.29	25.47
Married (%)	43.05	48.95	41.31	47.03	47.79	46.55
Primary or less (%)	39.05	31.78	41.20	20.25	19.74	20.57
Secondary (%)	37.47	43.39	35.72	38.33	40.19	37.14
High school (%)	18.30	19.33	17.99	27.30	26.64	27.72
College or more (%)	5.18	5.50	5.09	14.12	13.43	14.56
Panel C: Municipal Characteristics (Census 2010)						
Mean municipal population	14,912	12,355	15,779	87,360	107,973	78,390
Indigenous language speakers (%)	0.222	0.064	0.276	0.099	0.057	0.118
Rural population (%)	0.876	0.889	0.872	0.228	0.213	0.235
Infant population, 0–2 (%)	0.057	0.054	0.058	0.057	0.057	0.057
Child population, <12 (%)	0.260	0.245	0.265	0.250	0.254	0.249
Illiteracy rate, 15+ (%)	0.171	0.114	0.190	0.095	0.072	0.104
No access to piped water (%)	0.249	0.190	0.269	0.130	0.111	0.138
No health insurance (%)	0.386	0.332	0.405	0.362	0.338	0.372
No drainage/sanitation (%)	0.336	0.250	0.366	0.129	0.114	0.136
Panel C (cont.): Agriculture (SIAP 2009–2012)						
Avg. crop varieties per municipality	8.7	9.8	8.4	13.0	14.7	12.2
Avg. area sown per municipality (ha)	7,346	9,145	6,735	11,007	15,053	9,253
Rainfed share of area sown (%)	85.67	70.77	90.72	73.14	53.00	81.87
Avg. area sown, OI (ha)	829	977	779	2,553	4,580	1,674
Avg. area sown, PV (ha)	4,338	7,161	3,380	5,417	7,955	4,316
Avg. production value, OI (M pesos)	19.3	31.0	15.4	80.7	170.1	41.9
Avg. production value, PV (M pesos)	49.1	78.1	39.3	97.2	162.2	69.1

Notes: Panels A and B report individual-level means from SINAC birth records, 2009–2012. Panel C reports municipality-level means; census variables are from 2010, agriculture variables are within-municipality means over 2009–2012 then averaged across municipalities in each group. Dry = *Seco*; Non-dry = *Subhúmedo* + *Húmedo*. Rural municipalities defined as those with >50% of population in localities ≤2,500 inhabitants (Census 2010). Production values in millions of 2018 real pesos (M pesos).

4.2 Climate data

The main source for precipitation data is the Climate Hazards Group InfraRed Precipitation with Stations version 3 (CHIRPS v3). CHIRPS v3 provides quasi-global daily precipitation estimates at a 0.05° spatial resolution from 1981 to the present, combining remote-sensing data with in-situ station observations.[5] I aggregate daily precipitation to monthly municipality-level totals (in mm) within municipal boundaries. Temperature data come from the CHIRTS-ERA5 dataset, which provides daily mean temperature at the same spatial resolution.[6]

Similar to the literature on meteorological droughts (Rocha and Soares 2015; Lin et al. 2021; Edwards et al. 2024), I construct a measure of municipal specific deviations from historical patterns. Specifically, for each municipality j and birth month t, I compute the log average daily precipitation over the nine months preceding birth and compare it to the municipality-specific long-term geometric average over the same nine calendar months, using 1981–2008 as the baseline period. The measure is represented in equation (1):

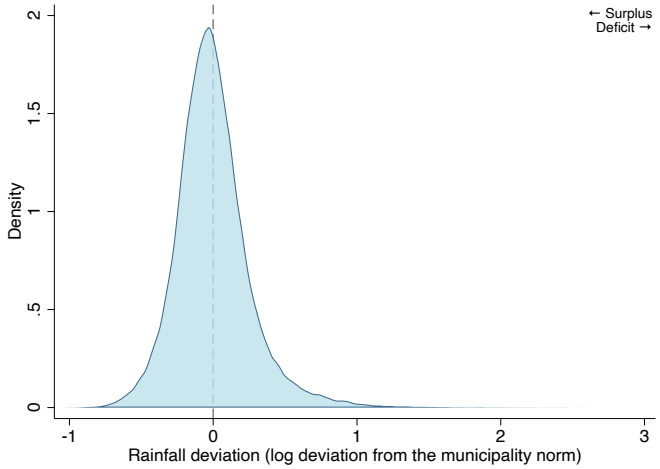
$$\Delta\text{RainDef}_{jt} = -\left[\ln\left(\frac{1}{9}\sum_{s=t-8}^{t}\bar{r}_{js}\right) - \ell^{LT}_{j,\mathcal{M}(t)}\right], \qquad \ell^{LT}_{j,\mathcal{M}(t)} = \frac{1}{|B|}\sum_{b\in B}\ln P_{jb,\mathcal{M}(t)}, \tag{1}$$

where $\bar{r}_{js}$ is mean daily precipitation in municipality j during month s, $\mathcal{M}(t)$ is the set of nine calendar months that define the gestational window, $P_{jb,\mathcal{M}(t)}$ is the average daily precipitation over those same nine calendar months in baseline year b, and $B = \{1981, \ldots, 2008\}$ is the set of baseline years. The baseline ends in 2008 so that the historical norm is constructed entirely from years preceding the 2009–2012 episode. The negative sign means that positive values correspond to rainfall deficits. Figure 3 shows the distribution of the rainfall deficit measure for the 2009–2012 study period.

[5] CHIRPS was developed through a collaboration between the United States Geological Survey and the Climate Hazards Center to support drought monitoring and early warning systems. Details at `https://www.chc.ucsb.edu/data/chirps3`.

[6] See `https://www.chc.ucsb.edu/data/chirts-era5`.

Figure 3: Distribution of the Log Rainfall Deficit Measure



Notes: Kernel density of the 9-month log-deviation rainfall deficit (ΔRainDef) across all municipality-months in the 2009–2012 study period. Positive values indicate below-normal rainfall (Drier conditions).

In Appendix A I show that this measure is equivalent to the recentering procedure in Borusyak and Hull (2023), which addresses omitted variable bias stemming from non-random exposure to exogenous shocks. In other words, although rainfall itself is exogenous, the deficit is not a shock but a function of the shock and of the municipality's climate, so that units differ systematically in how large a deficit they expect to register. With this recentering, the regression compares municipalities that received less rainfall than their own counterfactuals, rather than comparing dry places to wet ones, for instance. In this case, households sorting into locations on climate, geography, or any other predetermined characteristic does not threaten identification.

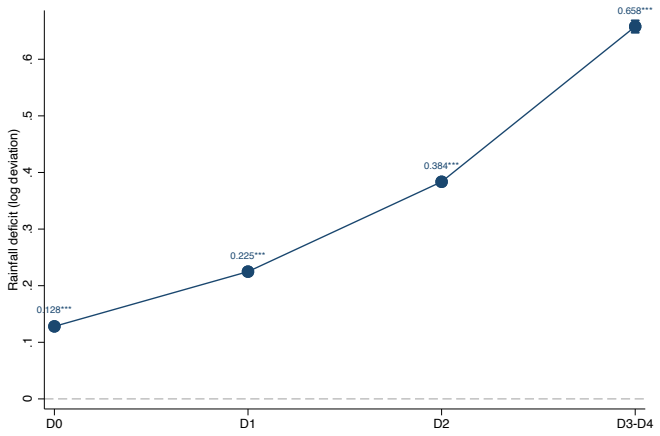
The log transformation allows direct interpretation in percentage terms and comparability across municipalities with different baseline precipitation levels.[7].

To verify that the rainfall deviation tracks actual drought conditions, I compare it against the official severity categories from CONAGUA's Monitor de Sequía de México (MSM), a composite index that combines seven meteorological, agricultural and hydrological indicators

[7]Since $\Delta\text{RainDef} = -\ln(P/P^{LT})$, the percentage reduction in rainfall relative to the historical norm is $1 - e^{-\Delta\text{RainDef}}$. For example, $\Delta\text{RainDef} = 0.25$ corresponds to a 22% reduction and $\Delta\text{RainDef} = 0.50$ to a 39% reduction. The raw standard deviation of the deficit arm is 0.249 log points, roughly a 22% rainfall reduction. Table B.1 reports how much of that variation remains under alternative fixed-effect structures.

with expert assessment.[8] Figure 4 plots the mean deviation by MSM category. The relationship is monotonic, with a steeper jump at D3–D4, so rainfall deficit deviations accurately correlate with the official classification. I use the continuous deviation as the main regressor because it isolates the meteorological component, has a direct percentage interpretation, and is comparable to measures used elsewhere in the literature.

Figure 4: Log Rainfall Deficit and MSM Drought Categories



Notes: Each point shows the mean value of the 9-month log-deviation rainfall deficit measure within the corresponding MSM drought category. Vertical bars show 95% confidence intervals.

4.3 Additional data

To study perinatal health consequences beyond birth outcomes, I use administrative discharge records from all public hospitals operated by the Ministry of Health (Secretaría de Salud), compiled by the Dirección General de Información en Salud (DGIS).[9] Each record includes patient sex, age, primary diagnosis coded under ICD-10, and municipality of residence. I aggregate discharges to the municipality-month level using SINAC births as the

[8]The MSM is part of the North American Drought Monitor (NADM), a collaboration between Mexico, the United States, and Canada. It classifies drought into five categories (D0 through D4) based on approximate return periods over a 100-year distribution. Table B.2 in the Appendix describes each category. For methodological details on the NADM, see droughtmonitor.unl.edu/nadm. For the Mexican implementation, see smn.conagua.gob.mx.

[9]Data available at dgis.salud.gob.mx. The dataset reports discharges from facilities operated by the Secretaría de Salud, identified through the CLUES national registry of health establishments. IMSS, ISSSTE, and private hospitals maintain separate reporting systems.

denominator. For maternal complications, I focus on obstetric codes (O00–O99) and separately on conditions related to preterm labor (O42, O47). For infant hospitalizations, I consider perinatal conditions (P00–P96) and congenital malformations (Q00–Q99). The panel covers 2010–2019 across 937 hospitals.

For exploring agricultural production as potential mechanism, I use annual agricultural production data from SIAP (*Servicio de Información Agroalimentaria y Pesquera*), which reports area sown, area harvested, production volume, and production value at the municipality-crop-year level.[10] I aggregate across annual crops (spring-summer and fall-winter cycles) and construct a balanced municipality-year panel for 2005–2012 covering 2,393 municipalities.

To assess air quality as a mechanism, I use satellite-derived $PM_{2.5}$ concentrations from the Atmospheric Composition Analysis Group (ACAG) V5.GL.05 dataset (van Donkelaar et al. 2021). The product combines aerosol optical depth from multiple satellite instruments with the GEOS-Chem chemical transport model, calibrated to ground-based stations. The native resolution is 0.01° (∼1 km). I aggregate to municipal monthly means using area-weighted averaging, generating a balanced panel of 2,450 municipalities covering the period of this study.

Municipal-level characteristics come from the 2010 Population Census produced by INEGI (Instituto Nacional de Estadística y Geografía). Table 1 reports these variables by urbanization and climate zone, including access to piped water, health insurance coverage, illiteracy rates, and the share of indigenous language speakers.

5 Empirical Strategy

The main specification estimates the effect of prenatal rainfall deviations on individual birth outcomes, allowing separate coefficients for a rainfall deficit and a rainfall surplus. For infant i born during month t to a mother residing in municipality j, the estimating equation is

[10] Data available at nube.siap.gob.mx/cierreagricola.

$$y_{ijt} = \beta_D \, \text{Deficit}_{jt} + \beta_S \, \text{Surplus}_{jt} + f(\text{Temp}_{jt}) + \mathbf{X}'_{ijt}\gamma + \alpha_{jq} + \delta_t + \theta_s \cdot t + \varepsilon_{ijt}, \tag{2}$$

where y_{ijt} is birth weight (in grams), an indicator for low birth weight, or an indicator for preterm birth. The two treatment terms are the size of the deviation on each side of the municipality norm, $\text{Deficit}_{jt} = |\Delta\text{RainDef}_{jt}| \cdot \mathbf{1}\{\Delta\text{RainDef}_{jt} > 0\}$ and $\text{Surplus}_{jt} = |\Delta\text{RainDef}_{jt}| \cdot \mathbf{1}\{\Delta\text{RainDef}_{jt} < 0\}$, with $\Delta\text{RainDef}_{jt}$ defined in equation (1). Both are non-negative, so a negative coefficient on birth weight means harm whether the deviation is dry or wet. [11]

$\mathbf{X}_{ijt}$ is a vector of maternal characteristics including marital status, parity, and years of schooling, and a quadratic in age. Following the climate literature, I also include a quadratic $f(\text{Temp}_{jt})$ in the 9-month average mean-centered temperature over the same gestational window (Deschênes et al. 2009; Barreca 2012; Hong 2025).

The fixed effects absorb variation at three levels. Municipality-by-calendar-quarter effects (α_{jq}) control for time-invariant municipal characteristics and seasonal patterns in both rainfall and birth outcomes within each municipality. Year effects (δ_t) absorb common annual shocks. State-specific linear trends ($\theta_s \cdot t$) allow for differential secular changes across states in both climate and health infrastructure. The coefficients are identified from within-municipality, within-quarter deviations in gestational rainfall from the local seasonal norm, after removing statewide trends. Standard errors are clustered at the municipality-by-quarter level to account for serial correlation within municipality-quarter cells across years.

The identifying assumption is that, conditional on the fixed effects and controls, deviations in gestational rainfall from the municipality-quarter norm are uncorrelated with unobserved determinants of birth outcomes. The recentered measure confines this restriction to the realization of the shock, so unobserved determinants of birth outcomes may differ arbitrarily across municipalities and no parallel-trends condition is imposed (See Appendix

[11] A single coefficient on $\Delta\text{RainDef}_{jt}$ imposes one straight line through the norm $\beta_D + \beta_S = 0$, and that restriction is rejected for all three outcomes as shown in the section of results.

A). The assumption also requires that rainfall in the gestational window be unanticipated. The nine-month averaging period is long enough that strategic timing of conception in response to anticipated rainfall is unlikely. Section B.4 further supports this with a placebo using rainfall leads and lags.

To study maternal and infant health consequences beyond birth, Section 8 estimates a municipality-month panel specification using the same fixed-effect structure. The dependent variables are hospital discharge rates and mortality rates per 1,000 live births in municipality j during month t. In this case regressions are weighted by annual births in the municipality to account for differences in cell size. Individual-level controls are not available in the hospitalization data, so the panel relies on the fixed effects and climate controls for identification.

6 Drought and Birth Outcomes

Table 2 presents estimates of prenatal rainfall deviations on birth outcomes. Panel A reports the two arms of the continuous measure. A one log point deficit reduces birth weight by 7.7 grams, raises the probability of low birth weight by 0.18 percentage points, and raises preterm birth by 0.30 percentage points, each significant at the one percent level. The surplus arm points the same way on all three outcomes, but in smaller magnitude and non distinguishable from zero.[12]

To put the magnitude of these coefficients in context, a severe deficit of 50% less rainfall than expected during gestation would reduce birth weight by 5.3 grams and increase the incidence of low birth weight and preterm delivery by 125 and 208 cases per 100,000 births, or about 2.2% and 3.4% above baseline.[13]

[12] An F test rejects linearity, $\beta_D + \beta_S = 0$, for the three outcomes ($p = 0.0004$, 0.009, 0.011). Under one straight line through the norm a surplus would improve birth outcomes by as much as an equally sized deficit worsens them. A second test does not reject symmetry, $\beta_D = \beta_S$, that a deviation of a given size matters equally whether it is dry or wet ($p = 0.096$, 0.812, 0.118).

[13] The India Meteorological Department classifies a seasonal rainfall deficit of more than 50% of the normal as severe drought (`https://mausam.imd.gov.in/imd_latest/monsoonfaq.pdf`). A 50% deficit corresponds to $\ln 2 \approx 0.693$ log points. Therefore, the effects are $7.715 \times 0.693 \approx 5.3$ grams for birth weight, and $0.0018 \times 0.693 \times 100{,}000 \approx 125$ and $0.0030 \times 0.693 \times 100{,}000 \approx 208$ cases per 100,000 births for low birth weight and preterm delivery, against baselines of 5,600 and 6,100.

Panel B conducts a first robustness exercise by replacing the continuous measure in equation 2 with indicator variables for the MSM drought categories (D0 through D4), as described in section 4.2. The results indicate that, relative to months classified as No Drought, the coefficients trace a monotonic and approximately linear dose-response relationship. D0 conditions reduce birth weight by 2.4 grams; D1, by 3.1 grams; D2, by 4.0 grams; D3–D4, by 5.9 grams. Low birth weight and preterm birth follow the same gradient. All coefficients are significant at conventional levels.

Table 2: Rainfall Deficits and Birth Outcomes

	Dependent Variable		
	Weight (g) (1)	LBW (2)	Preterm (< 37) (3)
Panel A. Rainfall deficit			
Deficit	-7.715***	0.002***	0.003***
	(1.375)	(0.001)	(0.001)
Surplus	-3.701	0.002	0.001
	(2.501)	(0.001)	(0.001)
Observations	7,470,462	7,470,462	7,470,462
Mean dep. var.	3163.262	0.056	0.061
Panel B. Drought categories			
D0	-2.415***	0.001***	0.001***
	(0.712)	(0.000)	(0.000)
D1	-3.127***	0.001**	0.001**
	(0.923)	(0.000)	(0.000)
D2	-4.035***	0.001**	0.002***
	(0.991)	(0.000)	(0.001)
D3–D4	-5.875***	0.001***	0.002***
	(1.201)	(0.000)	(0.001)
Observations	7,470,462	7,470,462	7,470,462
Mean dep. var.	3163.262	0.056	0.061

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Municipality × calendar-quarter FE, year FE, state-specific linear trends. Controls: mother's age, age^2, marital status, parity, schooling; mean-centered 9-month temperature and its square. SE clustered at municipality×quarter. Panel A: Deficit and Surplus are the two sides of the 9-month log-deviation, split at the municipality long-term norm, both non-negative. Panel B: drought-category indicators; omitted = No Drought.

I report additional series of robustness exercises in the Appendix. First, it is known that SINAC coverage is incomplete in its early years of operation and the shortfall concentrates in a few southern states (Mier y Terán Rocha and García Guerrero 2019). If registration itself

responds to drought, the estimates would reflect differential recording. Excluding the four states where SINAC captures less than 80 percent of the births in vital statistics does not change the main results (Table B.6). Second, a concern with the benchmark specification is that rainfall during gestation is correlated with rainfall in adjacent years, so that what I attribute to gestation may belong to those other periods. I therefore add the deviation of the same window one year before and one year after as additional controls (Table B.7). Neither lead term is statistically significant, as expected of rainfall that follows a birth already delivered. The lag is negative and significant at about three fifths of the contemporaneous magnitude, but smaller in magnitude, consistent with a persistent episode. Importantly, the main rainfall deficit coefficient remain largely unchanged.

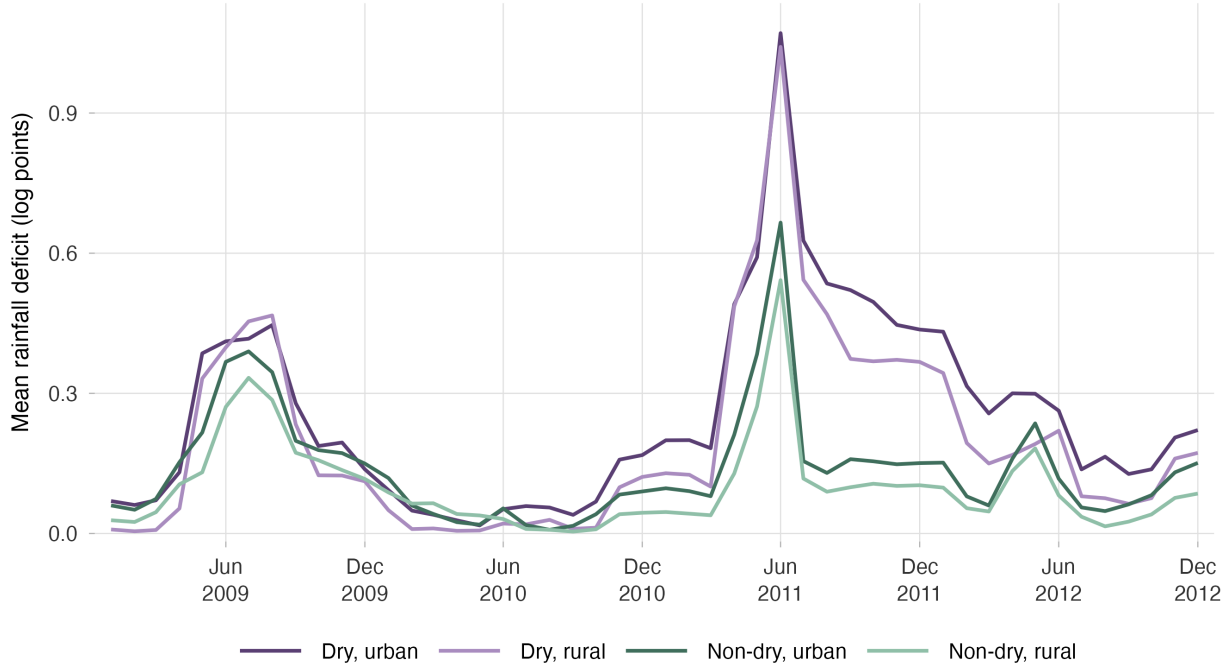
Third, one concern is that the estimates depend on the length of the exposure window. Table B.8 re-estimates the specification over three and six month windows. The effect on birth weight is indistinguishable from zero at three months and rises with each longer window. Shorter windows omit months of actual exposure, so this attenuation is what a cumulative effect over gestation would produce. Fourth, rainfall can be spatially correlated across neighbouring municipalities, which clustering at the municipality level treats as independent. Table B.9 recomputes the standard errors allowing for spatial dependence following Conley (1999), at cutoffs of 150 and 200 kilometres. The deficit coefficient remains significant at one percent for all three outcomes under both cutoffs.

Fifth, Figure C.1 plots the deficit coefficient across seven specifications. They range from state by quarter fixed effects to the preferred municipality by quarter structure, with a further variant that let the treatment interact flexibly with temperature. The deficit coefficient is insensitive to these choices. It stays negative and significant on all three outcomes in every specification, and closest together across the municipality-level rows. The temperature interactions are jointly zero, so the effect of a rainfall deficit does not vary with ambient temperature.

6.1 Heterogeneous effects across regions and urbanization

As Figure 1 shows, climate regions and socioeconomic conditions vary widely across Mexico. It is therefore reasonable to expect that regions responded differently to the 2011 shock. Additionally, Figure 5 shows that the deficit was not evenly distributed across climate regions or urbanization settings. In particular, dry areas moved furthest from their historical norm, though the deviation in non-dry areas was also large.

Figure 5: Birth-Weighted Rainfall Deficit by Climate Region and Urbanization



Notes: Birth-weighted municipal mean of Deficit, dated at the close of the 9-month window. Sources: CHIRPS v3; SINAC.

The rich SINAC administrative data cover the whole country, which allows me to exploit heterogeneity across climate regions and urbanization at national scale. Previous work restricts its samples to single geographies, whether the semiarid Northeast of Brazil (Rocha and Soares 2015), a single Chinese province (Lin et al. 2021), or purely rural regions in Sierra Leone (Abiona and Ajefu 2023).

Table 3 splits the sample by official climate classifications from CONAGUA into dry (arid and semi-arid) and non-dry municipalities (humid and sub-humid). Panel A shows

deficits reducing birth weight in both zones, by 7.3 grams in dry municipalities against 9.5 in non-dry, although the two are not statistically different. The incidence outcomes follow the same order. Low birth weight rises by 0.1 percentage points in dry municipalities and by 0.3 in non-dry ones. The corresponding figures for preterm birth are 0.2 and 0.4. The surplus arm is significant only in non-dry municipalities, at 6.6 grams, likely reflecting the intense hurricane season of 2010. In wetter municipalities, a deviation in either direction negatively affects birth weight. For comparison, the dry-zone point estimate is close in magnitude to the the 5.4 grams of birth weight per log point of rainfall reported in Rocha and Soares (2015) for the semiarid Northeast of Brazil under a similar identification strategy.

Panel B reports the dose-response through the MSM categories. Non-dry municipalities preserve the results reported for the whole sample, with birth weight falling from 3.1 grams at D0 to 7.7 grams at the more severe drought categories D3–D4. Similarly, low birth weight and preterm birth are positive at every category and largest at D3–D4. However, this gradient is not reproduced in dry municipalities. All three outcomes step up at D1 and stay flat through D3–D4, with birth weight settling between 4.7 and 5.1 grams. Low birth weight and preterm birth hold the same profile at positive values, though imprecisely estimated. In general, it is possible that the MSM

To further explore the heterogeneity of the results in terms of economic structure and urbanization within each climate zone, Table 4 disaggregates each climate region into rural and urban municipalities.[14]

I document three main patterns from these results. First, I observe statistically significant harmful effects of rainfall deficits on all three outcomes in non-dry municipalities. Rural areas within this climate region are particularly sensitive with birth weight falling by almost 22 grams per log point of rainfall deficit, close to three times the national estimate. This is consistent with the characteristics presented in Table 1 which shows that these municipalities

[14]INEGI classifies a locality as rural when it has fewer than 2,500 inhabitants. The locality is a finer geographic unit than the municipality. I therefore classify a municipality as rural when more than half of its population lives in rural localities.

Table 3: Rainfall Deficits and Birth Outcomes by Climate Zone

	Weight (g)		LBW		Preterm (< 37)	
	Dry	Non-Dry	Dry	Non-Dry	Dry	Non-Dry
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Rainfall deviation						
Deficit	-7.266***	-9.524***	0.001*	0.003***	0.002**	0.004***
	(2.141)	(1.849)	(0.001)	(0.001)	(0.001)	(0.001)
Surplus	-1.286	-6.637**	0.001	0.002	-0.000	0.002
	(4.039)	(3.122)	(0.002)	(0.001)	(0.002)	(0.002)
Observations	2,705,153	4,765,309	2,705,153	4,765,309	2,705,153	4,765,309
Mean dep. var.	3187.638	3149.425	0.055	0.057	0.063	0.060
Panel B. Drought categories						
D0	-1.124	-3.054***	0.000	0.001***	-0.000	0.002***
	(1.375)	(0.795)	(0.001)	(0.000)	(0.001)	(0.000)
D1	-5.114***	-1.844*	0.001**	0.000	0.001	0.001**
	(1.781)	(1.029)	(0.001)	(0.000)	(0.001)	(0.001)
D2	-4.727***	-3.721***	0.001	0.001**	0.001	0.002***
	(1.822)	(1.178)	(0.001)	(0.001)	(0.001)	(0.001)
D3–D4	-4.707**	-7.740***	0.001	0.002***	0.001	0.002**
	(1.952)	(1.617)	(0.001)	(0.001)	(0.001)	(0.001)
Observations	2,705,153	4,765,309	2,705,153	4,765,309	2,705,153	4,765,309
Mean dep. var.	3187.638	3149.425	0.055	0.057	0.063	0.060

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. All regressions include municipality × calendar-quarter FE, year FE, and state-specific linear trends. Individual controls: mother's age, age^2, marital status, parity, schooling; mean-centered 9-month temperature and its square. Standard errors clustered at municipality×quarter in parentheses. Panel A: Deficit and Surplus are the two sides of the 9-month log-deviation, split at the municipality long-term norm, both non-negative. Panel B: drought-category indicators; omitted = No Drought. Dry municipalities defined by Köppen–Geiger arid/semi-arid classification (BS/BW). Non-dry includes all other climate types. Assembled by hand from 01_bh_main_results_dry_muni_qrt and 01_bh_main_results_nondry_muni_qrt.

are the most deprived group on every measure, from access to piped water and sanitation to health insurance.[15] Furthermore, agriculture in these municipalities depends almost entirely on rainfall, which feeds 90.7% of the area sown against 70.8% in dry rural municipalities. Because of these reasons, these areas might be poorly adapted to rainfall deficits as those observed during our window of analysis.

Second, among dry municipalities the effect is present in urban areas, where the three

[15]For instance, although Chiapas has the largest renewable water availability per inhabitant in the country, it is also one of the poorest states, where infrastructure fails to deliver a daily supply of piped water, particularly in its rural areas (Soares 2007; Instituto Mexicano para la Competitividad 2023).

outcomes are close in magnitude to the national estimates. However, in dry rural areas I find no detectable effect on any of the three outcomes. One potential explanation of this result is that these regions are adapted to dry conditions, which is supported by the share of agricultural land that is irrigated, but also by the labor force composition and potential diversification of income. In dry rural areas 35% of the working population works in primary economic sectors versus 48% in non-dry.

However, the adaptation hypothesis is not totally convincing. As shown in Figure 5, the deviation of dry rural areas from their normal dry conditions was unusually severe. The hypothesis would also imply that dry urban areas, which have better infrastructure, are less affected, when in fact they are the ones where I find an effect within dry regions. Further, the drought coefficient on the incidence of low birth weight is negative, although imprecise, while the estimates for the other outcomes remain small. These pieces of evidence instead point to fetal selection in the SINAC records, since the data capture only live births and a shock that removes the weakest fetuses before delivery leaves the surviving newborns unrepresentative. Selection of this kind biases the estimated coefficients toward understating the true effect (Almond and Currie 2011). In Section 8 I show that this might be the cause in dry rural regions.

Third, birth weight and low birth weight respond similarly in urban areas of both climate regions, while the effect on preterm birth is more intense in non-dry urban municipalities, at 0.5 against 0.2 percentage points. Importantly, this group contains most of the Mexico City metropolitan area, the largest urban agglomeration in the country and among the most exposed to air pollution. Economic and medical literature relates air quality exposure during pregnancy to preterm delivery and to reductions in birth weight (Currie et al. 2009; Fong et al. 2019; Bello et al. 2026).

Table B.5 in the Appendix presents a fully interacted specification that estimates all four subgroup effects jointly in one regression and recovers the same qualitative pattern as the separate regressions of Table 4. The tests on the interaction terms show that non-dry rural

municipalities differs significantly from non-dry urban and from dry rural areas. The two urban groups are not distinguishable from each other on any outcome.[16]

These results suggest that outcomes react to drought differently across subregions. In particular, apart from the more direct channel of water availability, on which I have no data, the heterogeneity points to agricultural productivity, which can be important for rural areas in terms of income and nutrition, and to air quality deterioration, which meteorological literature has linked to dry conditions through the formation of dust and the prevention of wet deposition. These findings motivate the exploration of mechanisms in Section 7 and of other stages of perinatal health in Section 8.

[16]The fully interacted model regresses birth outcomes on the rainfall deficit and surplus, its interaction with a dry-climate indicator, its interaction with a rural indicator, and the triple interaction of all three. The recovered group-specific effects closely match the coefficients from Table 4. Formal tests of the linear combinations confirm that the subgroup gaps are statistically different for birth weight and low birth weight. I chose to present the separate regressions in the main text for clarity of interpretation.

Table 4: Rainfall Deficits and Birth Outcomes by Climate Region and Urbanization

	Weight (g)		LBW		Preterm (< 37)	
	Rural	Urban	Rural	Urban	Rural	Urban
Panel A: Dry municipalities						
Deficit	-2.2366	-7.8645***	-0.0018	0.0017**	0.0013	0.0020**
	(4.3792)	(2.3429)	(0.0021)	(0.0008)	(0.0024)	(0.0010)
Surplus	-3.8103	-1.2210	-0.0028	0.0015	-0.0077	0.0015
	(8.4501)	(4.4316)	(0.0040)	(0.0016)	(0.0051)	(0.0018)
Observations	299,790	2,405,363	299,790	2,405,363	318,388	2,541,336
Mean of Y (2009)	3188.20	3196.04	0.0498	0.0534	0.0623	0.0675
Panel B: Non-dry municipalities						
Deficit	-21.5135***	-7.1990***	0.0070***	0.0024**	0.0061***	0.0052***
	(4.4152)	(2.0166)	(0.0021)	(0.0009)	(0.0022)	(0.0016)
Surplus	-9.3794	-6.2355*	0.0025	0.0025	0.0046	0.0011
	(5.7318)	(3.7063)	(0.0026)	(0.0017)	(0.0032)	(0.0023)
Observations	999,576	3,765,733	999,576	3,765,733	1,076,753	4,010,168
Mean of Y (2009)	3184.20	3148.71	0.0430	0.0588	0.0498	0.0694

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Each cell is a separate regression on individual births, SINAC 2009–2012. Deficit and Surplus are the two sides of the 9-month BH log-deviation, split at the municipality long-term norm, both non-negative. Panel A restricts to dry-climate municipalities (Köppen–Geiger BS/BW); Panel B to non-dry. Rural is a municipal rural population share above 0.5. Municipality-quarter FE, year FE, state-specific linear trends. Controls: mother's age, age^2, marital status, parity, schooling; mean-centered 9-month temperature and its square. SE clustered at municipality-quarter in parentheses. Means computed for 2009 within each subsample. Assembled from 02_bh_rural_urban_dry_3col and _nondry_3col; 03h_bh_dnd_ru.do regenerates it directly.

6.2 Risk Components and Attributable Cases

Results in Table 4 are indicative of the sensitivity of the outcomes to deviations from normal rainfall conditions. However, they can lead to the misleading conclusion that non-dry rural areas are the most affected regions, since that is where the largest significant coefficients appear. In this section I show that understanding the burden of the shock also requires considering the geographical distribution of the event and the population at risk.

Following the insights of the disaster risk framework of the Intergovernmental Panel on Climate Change (IPCC), set out in its Special Report on Managing the Risks of Extreme Events and Disasters to Advance Climate Change Adaptation (SREX), risk (or burden) is a function of the interaction of at least three components (Cardona et al. 2012):

- **Hazard**. The occurrence of a physical event that may have adverse effects on the elements exposed to it.
- **Exposure**. The inventory of elements present in an area where such events may occur.
- **Vulnerability**. The propensity of those exposed elements to suffer adverse effects when the event occurs.

The framework defines risk over the distribution of hazards a geography may face. It has been applied mostly to comparisons across countries, where the available indicators restrict the analysis to the national level.

Under the same framework, we can conduct an ex post analysis and measure the burden over the hazard that a geography actually faced. Because our data allow us to estimate the sensitivity of at least four subregions, we can identify where the burden of the 2009–2012 drought falls and which of the three components drives it across the country. This distinction is policy relevant because a burden driven by hazard and one driven by vulnerability suggest different potential interventions. In particular, for municipality m, we can define the burden of the 2011 event as follows:

$$\text{Burden}_m = \underbrace{\hat{\beta}_{D,g(m)}}_{\textbf{Vulnerability}} \times \sum_t \underbrace{\text{Deficit}_{mt}}_{\textbf{Hazard}} \times \underbrace{\text{Births}_{mt}}_{\textbf{Exposure}}, \tag{3}$$

where $g(m)$ is the climate-by-urbanization group of municipality m and t indexes the months of 2011. The burden is accumulated month by month, since both the deficit and the number of births vary within the year. Only the deficit arm enters, so a month wetter than the municipality's own norm contributes zero.

Figure 6 presents a visual representation of this decomposition and the attributable burden of birth weight loss across the Mexican territory. Panels (a), (b) and (c) are the components of the burden, and panel (d) is the burden by municipality.[17]

The estimated burden indicates that in 2011 the toll of birth weight losses concentrated in the north of the country. It includes dry urban municipalities, but also sub-humid rural municipalities in the highlands of the Sierra Madre Occidental and Oriental. The national average loss is 2.4 grams per birth. In the worst municipalities of Chihuahua, Durango and Nuevo Leon it reaches 13 to 15 grams, where the highest vulnerability in the country met a severe deficit.

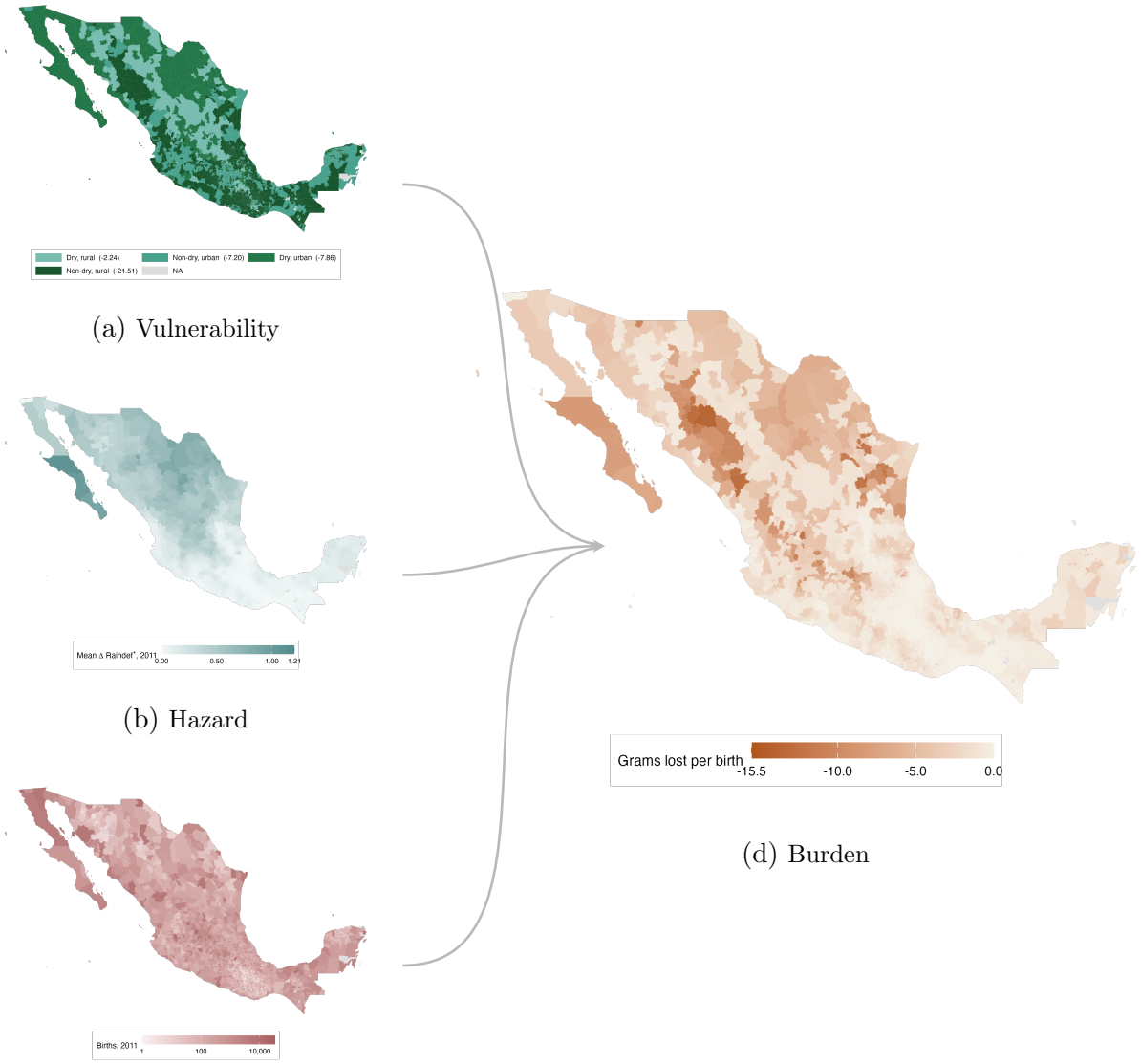
However, the same burden can reflect distinct causes. For instance, both Monterrey and a group of rural highland municipalities in the State of Mexico and Veracruz lost around 5 grams per birth. In Monterrey the loss comes mostly from the hazard ($7.9 \times 0.63 \approx 5$), while in the highlands it reflects mostly vulnerability ($21.5 \times 0.23 \approx 5$). In summary, neither component is sufficient on its own. The burden map is a relevant policy tool for understanding which interventions might work better in different regions.

Figure 7 repeats a similar exercise for preterm birth, with the burden expressed as a share of the cases each municipality would otherwise have expected.[18] The median municipality

[17] Panel (d) reports the burden per birth, $\text{Burden}_m / \sum_t \text{Births}_{mt}$. The panel therefore shows how much birth weight a typical newborn lost, and is comparable across municipalities of any size.

[18] Panel (d) reports $100 \times \text{Burden}_m / E_m$, where E_m sums national preterm rates by maternal age, parity and schooling over the municipality's own births. The denominator is the number of preterm births the municipality would record if it behaved like the national average given the mothers' characteristics, so the panel reads as the percentage the drought added to that baseline.

Figure 6: The 2011 Rainfall Deficit Burden on Birth Weight and its Risk Components



(a) Vulnerability

(b) Hazard

(c) Exposure

(d) Burden

Notes: Components of equation (3), 2011. Panel (a): $\hat{\beta}_{D,g(m)}$ from Table 4, grams per log point of rainfall deficit. Panel (b): mean monthly deficit against the municipality's own norm; surplus months enter as zero. Panel (c): live births, log scale. Panel (d): grams lost per birth, attributable to the deficit. Grey municipalities have no births in 2011. Non-dry rural differs from dry rural ($p = 0.001$) and from non-dry urban ($p = 0.004$); the remaining groups are not statistically separable.

records a burden equal to 0.7% of its expected preterm cases, and we observe a maximum of 6.8% mostly concentrated in the north eastern part of the country,

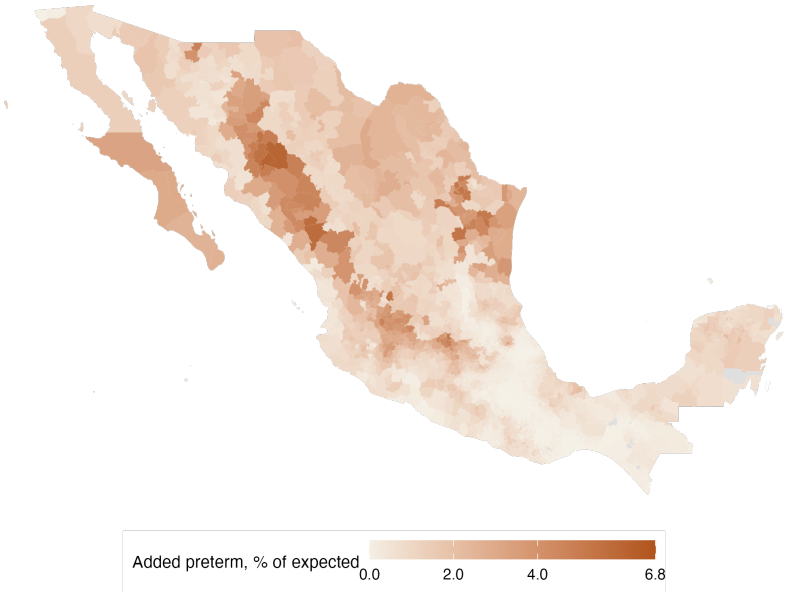
To give another perspective of these results, from the top 10% of most affected munic-

ipalities in terms of preterm burden, a substantial share clusters in the highlands of the Sierra Madre Occidental across Chihuahua and Durango, including the Sierra Tarahumara (Figure 7), where more than 80% of municipalities are classified as highly marginalized according to the official index of the National Population Council,[19] and where around 28% of the population is indigenous, against 6% nationally.

The remaining municipalities lie mainly in the northeastern states of Nuevo León, Tamaulipas and Coahuila, and in the centre-west and the Bajío. These include medium and large cities such as Torreón, Matamoros and Nuevo Laredo, along with several municipalities of the Monterrey metropolitan area. The contrast between the sierra municipalities and the rest of the top decile is instructive since, as in the case of birth weight, the same burden can reflect distinct causes. In the Sierras the burden comes mostly from its vulnerability, where the largest estimated coefficient of the rainfall deficit (Table 4) meets an ordinary deficit. Conversely, in the northern cities it comes mostly from the hazard, where a deficit twice as deep meets a coefficient a third the size. Neither component is sufficient on its own to understand the spatial distribution of the health damage.

[19] Índice de marginación, Consejo Nacional de Población, `https://www.gob.mx/conapo/documentos/indices-de-marginacion-2020-284372`. The index sorts municipalities into five degrees of marginalization, from very low to very high. Highly marginalized refers to the two highest.

Figure 7: Attributable Preterm Births as a Share of Expected, 2011



Notes: Attributable preterm cases as a percentage of expected cases, $100 \times \text{Burden}_m/E_m$, where E_m applies national preterm rates by maternal age, parity and schooling to the municipality's own births. Positive values are cases added. Grey municipalities have no births recorded in 2011.

7 Mechanisms

The fact that Section 6 documents heterogeneous effects across climate zones and urbanization, with different outcomes responding in different subgroups, points to multiple channels through which rainfall deficits can affect fetal and newborn health. The economics literature on drought and birth outcomes usually treats agricultural disruptions as the primary pathway. Drought reduces yields, raises local food prices, reduce real incomes, and affects maternal and infant nutrition. The assumption is common but seldom tested rigorously. Rocha and Soares (2015) for Brazil and Lin et al. (2021) for China both find suggestive evidence against the channel, yet neither study contrast it from competing pathways.

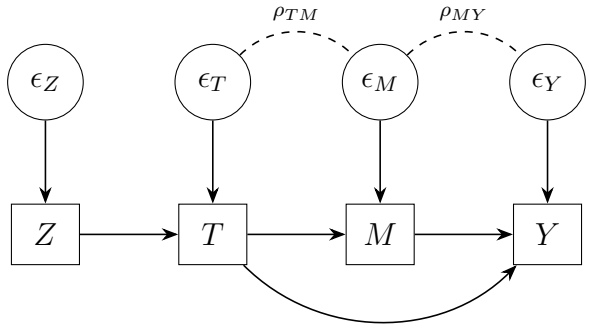
The air quality channel has received less attention in this literature, even though a separate body of research has documented its importance for birth outcomes (Currie et al. 2009). Indeed, drought raises particulate concentrations through wildfire ignition, soil desiccation, and reduced wet deposition (Ebi et al. 2021; Salvador et al. 2023), but no study has connected this pathway to fetal and newborn health in a drought context.

In this section I test both channels following the causal mediation model developed in Dippel et al. (2022), in the context of international trade and voting behaviours.

7.1 Conceptual framework

I consider a causal mediation model similar to Dippel et al. (2022). In the original partially confounded IV model the authors consider a problem in which both the main treatment T and the mediator M are endogenous (Figure 8). The model allows unobserved confounders that jointly affect the treatment and the mediator ($\rho_{TM} \neq 0$), as well as confounders that jointly affect the mediator and the outcome ($\rho_{MY} \neq 0$). However, it rules out confounding between the treatment and the outcome ($\rho_{TY} = 0$). Confounding is therefore partial, and this is the restriction that identifies the model.

Figure 8: The Partially Confounded IV Model



$$Z = \epsilon_Z$$

$$T = \beta_T^Z \, Z + \epsilon_T$$

$$M = \beta_M^T \, T + \epsilon_M$$

$$Y = \beta_Y^T \, T + \beta_Y^M \, M + \epsilon_Y$$

Notes: The model of Dippel et al. (2022). Boxes are observed variables and circles are error terms, with $\epsilon_Z, \epsilon_T, \epsilon_M, \epsilon_Y$ of variances $\sigma^2_{\epsilon_Z}, \sigma^2_{\epsilon_T}, \sigma^2_{\epsilon_M}, \sigma^2_{\epsilon_Y}$. Solid arrows are structural relations and the curved arrow from T to Y is the direct effect β_Y^T. Dashed arcs denote correlated errors: $\rho_{TM} \neq 0$ and $\rho_{MY} \neq 0$ are permitted, while the absence of an arc between ϵ_T and ϵ_Y is the identifying restriction $\rho_{TY} = 0$.

The objects of interest are the direct effect $DE = \beta_Y^T$, the indirect effect $IE = \beta_M^T \cdot \beta_Y^M$, and the total effect $TE = DE + IE$.[20] Dippel et al. (2022) show that under this restriction a single instrument identifies the four structural coefficients, and therefore the full decomposition.

The case of this paper is simpler. Here T is the municipality-specific meteorological rainfall deficit Deficit_{mq} and the outcome Y is a birth outcome. Following the weather shocks literature, I treat the deficit as exogenous, so that conditional on the geographical and seasonal fixed effects and the demographic and municipal controls it is uncorrelated with the unobserved determinants of birth outcomes and $\rho_{TY} = 0$ holds by construction. Therefore, only the mediator is endogenous, and identifying its coefficient requires an excluded instrument distinct from the treatment.

With a valid instrument Z_{mq} for the mediator, the model can be estimated as follows:

$$M_{mq} = \pi_0 \, \text{Deficit}_{mq} + \pi_S \, \text{Surplus}_{mq} + \pi_1 \, Z_{mq} + \mathbf{X}_i' \delta + f(\text{Temp}_{mq}) + \alpha_{mq} + \gamma_t + \delta_s \cdot t + u_{mq}$$

$$y_i = \beta_T \, \text{Deficit}_{mq} + \beta_S \, \text{Surplus}_{mq} + \beta_M \, \widehat{M}_{mq} + \mathbf{X}_i' \delta + f(\text{Temp}_{mq}) + \alpha_{mq} + \gamma_t + \delta_s \cdot t + \varepsilon_i \tag{4}$$

[20] The total effect is the derivative $dY/dT = \underbrace{\partial Y/\partial T}_{DE = \beta_Y^T} + \underbrace{(\partial M/\partial T)(\partial Y/\partial M)}_{IE = \beta_M^T \beta_Y^M}$.

with controls and fixed effects as in the main specification. The surplus arm enters both stages as a control and is not reported, so the decomposition refers throughout to the deficit. The direct effect is β_T, the indirect effect is $\beta_M \pi_0$, and their sum is the total effect. The exclusion restriction is that Z_{mq} affects birth outcomes only through M_{mq}, conditional on both arms and the controls.[21]

7.2 Agricultural productivity

Agricultural productivity can affect fetal and newborn health through several channels. On the one hand, if rainfall deficits reduce the harvest, they affect the income of families that rely on agricultural labor while pushing up prices in local markets that are not connected to international prices. On the other hand, families that depend on subsistence farming may lose a direct source of nutrition (Arceo-Gómez et al. 2020). Additionally, both paths can increase maternal stress, which has been documented to have detrimental effects on the mother and the fetus during gestation (Vins et al. 2015).

The channel is most plausible in municipalities with intensive rainfed agriculture, where rainfall variation maps directly into agricultural outcomes variation. The median rainfed share of area sown in the sample was about 80% in the period analyzed, but it varies across subgroups, from 53% in dry urban municipalities to 91% in non-dry rural ones (Table 1).

I estimate the causal mediation model 4 using the crop loss rate reported in each municipality during 2009 to 2012. I instrument this potential mediator using the rainfall deficit interacted with the municipality's pre-period exposure to rainfed cultivation. Spoecifically,

$$Z_{mq} = \text{Deficit}_{mq} \times \left(\tilde{R}_m - \bar{R} \right), \tag{5}$$

where $\tilde{R}_m$ is the share of area sown under rainfed cultivation in municipality m over 2005–

[21] Substituting the first stage into the second and collecting terms in the deficit gives the reduced form $y_i = (\beta_T + \beta_M \pi_0)\,\text{Deficit}_{mq} + (\beta_M \pi_1)\, Z_{mq} + \cdots$. Writing the two reduced-form coefficients as θ_T and θ_Z and inverting, $\beta_M = \theta_Z / \pi_1$ and $\beta_T = \theta_T - \beta_M \pi_0$. The mediator coefficient is a ratio of the two Z coefficients, the reduced form over the first stage, and the deficit coefficient is what the level term retains once that ratio is imposed.

2008 and $\bar{R}$ is its mean in the estimation sample. The share is predetermined with respect to the drought episode, and it scales how much of the municipality's crops are exposed to a rainfall deficit.

The identification assumption is that rainfed exposure modifies the effect of the rainfall deficit on birth outcomes only through agricultural production. In other words, for the instrument to fail it must be that drought affects birth outcomes differentially by rainfed exposure through a channel unrelated to agricultural productivity, conditional on the controls and the remaining interactions. [22]

Table 5 presents the results of this exercise. The first column shows that both the main deficit variable Deficit and the interaction as excluded instrument induce crop losses. In other words, while rainfall deficits increase crop losses, they do it more in areas that previous to the shock depended more on rainfall. The subsequent three columns show the second stage. We can observe that the direct effect of Deficit is close to the reduced form I presented in the main results in Table 2. However, the second stage coefficients for Crop Losses are small and not statistically significant. This means that with this design, birth outcomes are not affected by the crop losses induced by the droughts, and crop losses are not a mediator with an independent indirect effect.[23]

As shown in the descriptive statistics of Table 1, there is variation in the share of rainfed agriculture across different climatic regions, so the instrument is not equally strong across them. In historically non-dry municipalities cultivation was almost universally rainfed, with too little variation for the instrument to pick up. Conversely, in historically dry areas rainfed agriculture represents 64.7% of the area sown, with enough variation.[24]

[22]Both arms of the deviation are also interacted with the rural population share, log population, the dry-climate indicator and a deprivation index, built as the first principal component of seven census indicators including illiteracy, dirt floors and the lack of piped water, sanitation and health insurance. The excluded instrument does not pick up differential effects of drought by poverty, remoteness or size

[23]Tables B.10, B.11, and B.12 in the appendix report the OLS estimates for each outcome. The reduced form of the instrument on the outcomes is not statistically significant, which means that birth outcomes are not differentially affected in areas with previous higher dependence on rainfed agriculture. Adding crop losses to the reduced form is also not statistically significant. The inclusion of both variables in the OLS does not affect the magnitude or the significance of the direct effect of Deficit.

[24]In non-dry municipalities the pre-period rainfed share averages 90.0% with a standard deviation of 17.3,

Table 5: Drought, Crop Loss and Birth Outcomes

	First stage	IV second stage		
	(1)	(2)	(3)	(4)
Dep. variable	CL	BW	LBW	PT
Deficit (direct)	8.458***	-10.940***	0.0026***	0.0037***
	(1.232)	(1.916)	(0.0007)	(0.0008)
Deficit × Pre-Rainfed	0.315***			
	(0.052)			
Crop loss (%)		0.147	0.00000	-0.00001
		(0.135)	(0.00005)	(0.00006)
Indirect effect		1.241	0.0000	-0.0001
		(1.145)	(0.0004)	(0.0005)
Total effect		-9.699***	0.0026***	0.0036***
		(1.516)	(0.0007)	(0.0008)
KP F-statistic	37.07			
AR p ($\beta = 0$)		0.311	0.996	0.866
Observations	6,665,565	6,665,565	6,665,565	6,665,565

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012. CL: crop loss rate (%), SIAP municipality-year, instrumented in (2)–(4) with Deficit × Pre-Rainfed. Deficit: 9-month BH rainfall deficit. Pre-Rainfed: 2005–08 rainfed share of sown area, centred at 72.72%. Both arms interacted with rural share, log population, dry indicator and a deprivation index; surplus arm and these interactions unreported. IE = CL coefficient × first-stage Deficit. TE = direct + indirect. KP F: Kleibergen-Paap rk Wald. AR: Anderson-Rubin test of $\beta = 0$. FE: municipality × quarter, year, state trends. Controls: maternal age, age^2, marital status, parity, schooling, temperature quadratic. SE clustered by municipality.

Table 6 presents the results for both climate zones. Panel A reports historically dry municipalities. Similar to the full sample result, the first stage is strong and the second stage is small in magnitude and insignificant. In dry areas we can conclude that crop losses are not a mediator from drought to birth outcomes. Panel B reports historically non-dry municipalities. Although drought does affect crop losses, these are not differentially affected in land that was more rain dependent at the baseline, meaning that in these areas the instrument does not have a first stage. In non-dry municipalities this design is inconclusive.

I check the robustness of the results using an alternative instrument and an alternative

a tenth percentile of 64.2 and a median of 98.4, and 5.0% of them are majority-irrigated. In dry municipalities the mean is 64.7% with a standard deviation of 35.2, a tenth percentile of 1.4 and a median of 79.3, and 29.9% are majority-irrigated. Identification requires an irrigated comparison group, so it is the dispersion of the rainfed share and not its level that gives the instrument its strength.

Table 6: Agricultural Mediation by Climate Zone, Individual Births

	(1)	(2)	(3)	(4)
	First stage	IV second stage		
Dep. variable	CL	BW	LBW	PT
Panel A. Dry municipalities				
Deficit	7.498***	-7.877**	0.0007	0.0019**
	(1.748)	(3.367)	(0.0009)	(0.0009)
Deficit × Pre-Rainfed	0.297***			
	(0.065)			
Crop loss (%)		0.086	0.00003	-0.00000
		(0.187)	(0.00006)	(0.00007)
KP F	21.13			
Indirect effect		0.643	0.0002	-0.0000
Total effect		-7.233	0.0009	0.0019
Observations		2,518,697		
Panel B. Non-dry municipalities				
Deficit	7.214***	-11.510	-0.0032	-0.0024
	(1.631)	(8.105)	(0.0077)	(0.0077)
Deficit × Pre-Rainfed	0.071			
	(0.071)			
Crop loss (%)		-0.097	0.00102	0.00103
		(1.184)	(0.00119)	(0.00115)
KP F	1.01			
Indirect effect		-0.703	0.0073	0.0074
Total effect		-12.213	0.0041	0.0050
Observations		4,146,868		

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012, panels estimated separately by CONAGUA climate zone. CL: crop loss rate in percent, SIAP, municipality-year. Deficit: size of the 9-month BH rainfall deficit; in the IV columns its coefficient is the direct effect. Columns (2)–(4) instrument CL with Deficit × Pre-Rainfed. The surplus arm and its Pre-Rainfed interaction enter every column as controls, not reported. Pre-Rainfed: 2005–2008 rainfed share of area sown, centred within panel at 50.95 (dry) and 85.94 (non-dry) percent. All columns interact both arms with rural share, log population, and a deprivation index, re-centred within panel; the dry indicator is constant within panel and drops. Indirect effect: CL coefficient × first-stage Deficit; total = direct + indirect. FE: municipality × quarter, year, state trends. Controls: maternal age and its square, marital status, parity, schooling, temperature quadratic. SE clustered by municipality.

mediator. Table B.13 of the appendix reports the same exercise instrumenting crop losses with the interaction between the rainfall deficit and the baseline share of area sown in the spring-summer cycle, which is the cycle that mostly depends on rainfall. The first stage

is strong, with an F-statistic of 29.6, and the coefficient on crop losses remains small and statistically insignificant.

Table B.14 instruments crop losses with both interactions at once. Each enters the first stage significantly, so the two provide separate variation in the mediator. The second stage remains null. The Hansen J test does not reject the overidentifying restriction for any of the three outcomes ($p = 0.74$, 0.65 and 0.61), which means that the two instruments deliver statistically indistinguishable estimates of the mediator coefficient, conditional on one of them being valid.[25]

Table B.15 uses the volume of production as mediator instead of crop losses. As before, the first stage is strong and significant, meaning that drought reduces the volume of production and does it more in initially more rainfed areas, while the second stage is null and small for the agricultural mediator. The indirect effect in all the exercises is of a similar magnitude and statistical non significance.

It is important to note that, under this causal mediation design, the results depend on the choice of instrument. Each interaction identifies the effect of agricultural outcomes only where pre-period exposure varies, and it recovers an average weighted towards those municipalities. The conclusion then is not that agriculture does not mediate the effect of drought on health, but that under these instruments its contribution is small relative to the total effect of the drought. However, since drought unambiguously affects agricultural production importantly in all regions, a contribution this small is surprising. In the following section I bound the largest mediation effect the agricultural channel could have, with a design that answers a slightly different question, namely how large the effect of the agricultural mediator on birth outcomes would have to be for that channel to account for the whole effect of the deficit.

[25]The test cannot say which of the two is valid. Its power also depends on the second instrument moving the mediator once the first is included, and the conditional F-statistic of the spring-summer interaction is 4.25 against 16.9 for the rainfed share.

7.3 Air quality. $PM_{2.5}$ concentrations

Particulate exposure can affect fetal and newborn health through a more direct physiological route, since inhaled pollutants induce inflammatory and prothrombotic processes with documented effects on birth weight and gestation (Salvador et al. 2023; Currie et al. 2009; Arceo et al. 2016). On the other hand, the climatological literature relates rainfall deficits to particulate concentrations, since drought and high winds mobilise dust and raise the particulate matter suspended in the atmosphere (Ebi et al. 2021).

To estimate the potential mediation of the drought effect by air quality in model 4, I use satellite-derived $PM_{2.5}$ concentrations from the Atmospheric Composition Analysis Group (ACAG) V5.GL.05 dataset (van Donkelaar et al. 2021). Since the channel is most plausible in municipalities with a dense vehicle fleet, I define the excluded instrument as the rainfall deficit interacted with the municipality's vehicle fleet,

$$Z_{mq} = \text{Deficit}_{mq} \times \left(\ln C_m - \overline{\ln C}\right), \tag{6}$$

where C_m is cars per 1,000 residents. Drought dries road surfaces and the surrounding soil, and the resuspension of that material scales with local traffic, so a rainfall shortfall raises particulate concentrations more where the passenger fleet is denser.

Table 7 presents the results. The first stage indicates that a one log point rainfall deficit increases ambient $PM_{2.5}$ by 0.73 $\mu g/m^3$, and that the response is larger in municipalities with a denser car fleet, which is the identifying variation of the excluded instrument. In the second stage, the direct effect of rainfall deficit is close to the reduced form of the main results, while the coefficient on $PM_{2.5}$ is small and not statistically significant for the three outcomes. The implied indirect effect is below one gram of birth weight. [26]

[26]Tables B.16, B.17 and B.18 in the appendix report the OLS estimates. The reduced form of the instrument is not statistically significant, so birth outcomes are not differentially affected in municipalities with a denser car fleet. The OLS coefficient on $PM_{2.5}$ is 0.278 grams per $\mu g/m^3$ and significant, but ambient concentrations are correlated with unobserved determinants of birth outcomes, so it is not informative about exposure. Neither variable moves the direct effect of Deficit.

Table 7: Drought, Air Quality and Birth Outcomes

	First stage	IV second stage		
	(1)	(2)	(3)	(4)
Dep. variable	$PM_{2.5}$	BW	LBW	PT
Deficit (direct)	0.734***	-9.364***	0.0029***	0.0037***
	(0.147)	(2.605)	(0.0010)	(0.0013)
Deficit × Log cars	1.178***			
	(0.236)			
$PM_{2.5}$		0.772	-0.00054	-0.00010
		(3.030)	(0.00112)	(0.00136)
Indirect effect		0.567	-0.0004	-0.0001
		(2.225)	(0.0008)	(0.0010)
Total effect		-8.797***	0.0025***	0.0037***
		(1.368)	(0.0006)	(0.0008)
KP F-statistic	24.97			
AR p ($\beta = 0$)		0.799	0.632	0.941
Observations	7,262,572	7,262,572	7,262,572	7,262,572

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012. $PM_{2.5}$: municipal mean in $\mu g/m^3$, month of birth, instrumented in (2)–(4) with Deficit × Log cars. Deficit: 9-month BH rainfall deficit. Log cars: 2009 registered cars per 1,000 residents, in logs, centred at 4.896. Bus, freight, motorcycle and public-service rates enter on the same scale as controls. Both arms interacted with rural share, log population, dry indicator and a census deprivation index; surplus arm and these interactions unreported. IE = $PM_{2.5}$ coefficient × first-stage Deficit. TE = direct + indirect. KP F: Kleibergen-Paap rk Wald. AR: Anderson-Rubin test of $\beta = 0$. FE: municipality × quarter, year, state trends. Controls: maternal age, age^2, marital status, parity, schooling, temperature quadratic. SE clustered by municipality × quarter.

Because the density of the car fleet differs between rural and urban areas, the instrument is not equally strong in both. Table 8 reports the causal mediation model estimated separately for each. In urban municipalities, Panel A, a one log point rainfall deficit increases $PM_{2.5}$ by 0.83 $\mu g/m^3$, with a larger response where the fleet is denser. In rural municipalities, Panel B, the deficit on its own is 0.08 and not statistically significant, while the interaction reaches 1.74 against 0.91 in cities. Rural air quality responds to drought only where there are vehicles, with a steeper response than in cities because the fleet starts from a much lower base. The instrument is stronger in the rural panel, with a Kleibergen-Paap statistic of 40.6 against 11.1 in urban areas.

The coefficient on $PM_{2.5}$ is small and not statistically significant in both panels. The

direct effect of the deficit is −7.3 grams in urban municipalities and −17.7 in rural ones, in line with the heterogeneity reported in Table 4.

Table 8: Pollution Mediation by Urbanization, Individual Births

	(1)	(2)	(3)	(4)
	First stage	IV second stage		
Dep. variable	$PM_{2.5}$	BW	LBW	PT
Panel A. Urban municipalities				
Deficit	0.830***	-7.315*	0.0023	0.0035*
	(0.176)	(4.062)	(0.0015)	(0.0019)
Deficit × Log cars	0.908***			
	(0.273)			
$PM_{2.5}$		0.201	-0.00032	-0.00020
		(4.705)	(0.00161)	(0.00202)
KP F	11.06			
AR p		0.966	0.841	0.921
Indirect effect		0.167	-0.0003	-0.0002
Total effect		-7.149***	0.0021***	0.0034***
Observations		6,003,204		
Panel B. Rural municipalities				
Deficit	0.080	-17.733***	0.0056***	0.0049***
	(0.178)	(3.718)	(0.0017)	(0.0018)
Deficit × Log cars	1.738***			
	(0.273)			
$PM_{2.5}$		-1.301	-0.00034	0.00124
		(4.513)	(0.00223)	(0.00229)
KP F	40.63			
AR p		0.773	0.878	0.588
Indirect effect		-0.104	-0.0000	0.0001
Total effect		-17.837***	0.0056***	0.0050***
Observations		1,259,368		

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012, panels estimated separately. Rural: rural population share above 0.5. $PM_{2.5}$: municipal mean in $\mu g/m^3$, month of birth, instrumented in (2)–(4) with Deficit × Log cars. Deficit: 9-month BH rainfall deficit; in the IV columns its coefficient is the direct effect. Log cars: 2009 registered cars per 1,000 residents, in logs, centred within panel at 5.137 (urban) and 3.745 (rural). Bus, freight, motorcycle and public-service rates enter on the same scale as controls. Both arms interacted with rural share, log population, dry indicator and a census deprivation index, re-centred within panel; surplus arm and these interactions unreported. IE = $PM_{2.5}$ coefficient × first-stage Deficit. TE = direct + indirect. KP F: Kleibergen-Paap rk Wald. AR: Anderson-Rubin test of $\beta = 0$. FE: municipality × quarter, year, state trends. Controls: maternal age, age^2, marital status, parity, schooling, temperature quadratic. SE clustered by municipality × quarter.

7.4 Bounding the potential mediator effects

In this section I return to agricultural outcomes and air quality with a question independent of the previous instruments: What would the effect of the mediator have to be for the channel to account for the drought effect on its own? To answer it I restrict the mediation model of equation (4) by instrumenting the mediator with the rainfall deficit itself and removing the deficit from the second stage, so its direct effect is constrained to zero.[27] Provided the other non-included potential mediators do not improve birth outcomes, this estimate is an upper bound of the potential channel.[28] Therefore, the resulting coefficients can be compared with the per-unit effects the literature reports for each mediator.

Table 9 reports the exercise for crop loss. For the channel to account for the deficit effect on its own, crop loss would have to reduce birth weight by 0.71 grams per percentage point, so a municipality losing its entire harvest would record a loss of about 71 grams per birth, on average. Similarly, Table 10 reports the exercise for air quality. For the channel to account for the deficit effect on its own, $PM_{2.5}$ would have to reduce birth weight by 12.4 grams per $\mu g/m^3$.

To put these magnitudes in context, crop losses would have needed to rise by about 140 percent from their 2010 level for agriculture to account for the whole drought effect, while ambient $PM_{2.5}$ would have needed to rise by about 6 percent.[29] In proportional terms, the agricultural channel needs a large movement to produce what a much smaller air quality deterioration produces.

[27] Setting $\beta_T = 0$ in the reduced form $\theta_T = \beta_T + \beta_M \pi_0$ gives $\beta_M = \theta_T / \pi_0$.

[28] Write the reduced form as $\theta_T = \beta_M \pi_0 + \delta$, where δ collects every other route from the deficit to the outcome. The calibration returns $\theta_T / \pi_0 = \beta_M + \delta / \pi_0$, so it overstates the magnitude of β_M whenever δ has the sign of harm.

[29] A one log point deficit raises crop losses by 9.8 percentage points and $PM_{2.5}$ concentrations by 0.61 $\mu g/m^3$, against 2010 means of 7.0 percent and 10.5 $\mu g/m^3$.

Table 9: Crop Loss under Full Mediation

	(1)	(2)	(3)	(4)
	First stage	Full mediation		
Dep. variable	CL	BW	LBW	PT
Reduced form				
Deficit		-7.009***	0.0006	0.0017***
		(1.601)	(0.0005)	(0.0006)
Panel A. Full mediation, instrument Deficit				
Deficit	9.811***			
	(1.555)			
Crop loss (%)		-0.714***	0.0001	0.0002**
		(0.196)	(0.0001)	(0.0001)
KP F	39.80			
Observations		6,665,565		

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012, on the births where crop loss is observed. CL: crop loss rate (%), SIAP municipality-year. Deficit: 9-month BH rainfall deficit. Crop loss is instrumented with Deficit, which therefore does not enter the second stage. The coefficient is what the per-unit effect would have to be for the channel to account for the deficit effect alone. The model is just identified, so the calibration equals the reduced form divided by the first stage exactly. The surplus arm is a control in every regression, not reported. No moderator interactions enter: they are functions of the instrument. KP F: Kleibergen-Paap rk Wald. FE: municipality × quarter, year, state trends. Controls: maternal age, age^2, marital status, parity, schooling, temperature quadratic. SE clustered by municipality.

Table 10: Air Quality under Full Mediation

	(1)	(2)	(3)	(4)
	First stage	Full mediation		
Dep. variable	$PM_{2.5}$	BW	LBW	PT
Reduced form				
Deficit		-7.571***	0.0017***	0.0029***
		(1.376)	(0.0005)	(0.0007)
Panel A. Full mediation, instrument Deficit				
Deficit	0.612***			
	(0.052)			
$PM_{2.5}$		-12.366***	0.0028***	0.0047***
		(2.489)	(0.0009)	(0.0011)
AR p ($\beta = 0$)		0.000	0.001	0.000
KP F	137.12			
Observations		7,262,572		

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012. $PM_{2.5}$: municipal mean in $\mu g/m^3$, 9-month mean over gestation. Deficit: the deficit arm of the 9-month BH deviation. $PM_{2.5}$ is instrumented with the rainfall arms, which therefore do not enter the second stage. The coefficient is what the per-unit effect would have to be for air quality to account for the deficit effect alone. The surplus arm is a control in every regression, not reported. The model is just identified, so the calibration equals the reduced form divided by the first stage exactly. No moderator or fleet interactions enter: they are functions of the instrument. KP F: Kleibergen-Paap rk Wald. AR: Anderson-Rubin, weak-instrument robust. FE: municipality × quarter, year, state trends. Controls: maternal age, age^2, marital status, parity, schooling, temperature quadratic. SE clustered by municipality × quarter.

Table 11: Crop Loss under Full Mediation, by Climate Zone and Urbanization

	First stage		Full mediation			
	CL	KP F	BW	LBW	PT	Observations
Dry, rural	4.016***	7.01	-0.479	-0.0005	0.0007	299,114
	(1.517)		(1.144)	(0.0005)	(0.0005)	
Dry, urban	3.946**	4.67	-2.081*	0.0003	0.0005*	2,219,583
	(1.826)		(1.250)	(0.0002)	(0.0003)	
Non-dry, rural	10.133***	26.09	-2.137***	0.0007***	0.0005**	999,412
	(1.984)		(0.639)	(0.0002)	(0.0002)	
Non-dry, urban	5.299***	8.05	-1.613**	0.0005**	0.0009***	3,147,456
	(1.867)		(0.704)	(0.0002)	(0.0003)	

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012, each cell estimated separately. Dry follows the CONAGUA classification; rural is a rural population share above 0.5. CL: crop loss rate (%), SIAP municipality-year. Column (1) is the effect of the 9-month BH rainfall deficit on crop loss. Columns (3)–(5) instrument crop loss with that deficit, which therefore does not enter the second stage, and report what the per-unit effect would have to be for the channel to account for the deficit effect in that cell alone. Each cell is just identified, so the coefficient equals its reduced form divided by column (1). A cell with a first stage near zero returns a large coefficient for that reason. The surplus arm is a control in every regression, not reported. KP F: Kleibergen-Paap rk Wald. FE: municipality × quarter, year, state trends. Controls: maternal age, age^2, marital status, parity, schooling, temperature quadratic. SE clustered by municipality.

Table 12: Air Quality under Full Mediation, by Climate Zone and Urbanization

	First stage		Full mediation			
	$PM_{2.5}$	KP F	BW	LBW	PT	Observations
Dry, rural	0.565***	81.43	-4.134	-0.0033	0.0051	299,482
	(0.063)		(7.689)	(0.0037)	(0.0037)	
Dry, urban	0.725***	86.38	-10.259***	0.0019*	0.0023*	2,289,932
	(0.078)		(3.398)	(0.0010)	(0.0013)	
Non-dry, rural	0.461***	84.97	-45.658***	0.0147***	0.0111**	959,886
	(0.050)		(10.734)	(0.0048)	(0.0048)	
Non-dry, urban	0.711***	77.00	-10.093***	0.0035***	0.0059***	3,713,272
	(0.081)		(3.178)	(0.0013)	(0.0018)	

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012, each cell estimated separately. Dry follows the CONAGUA classification; rural is a rural population share above 0.5. $PM_{2.5}$: municipal mean in $\mu g/m^3$, 9-month mean over gestation. Column (1) is the effect of the 9-month BH rainfall deficit on $PM_{2.5}$. Columns (3)–(5) instrument $PM_{2.5}$ with that deficit, which therefore does not enter the second stage, and report what the per-unit effect would have to be for air quality to account for the deficit effect in that cell alone. Each cell is just identified, so the coefficient equals its reduced form divided by column (1). A cell with a first stage near zero returns a large coefficient for that reason. The surplus arm is a control in every regression, not reported. KP F: Kleibergen-Paap rk Wald. FE: municipality × quarter, year, state trends. Controls: maternal age, age^2, marital status, parity, schooling, temperature quadratic. SE clustered by municipality × quarter.

8 Consequences during pregnancy

A recurrent concern in the literature on in-utero exposures is fetal selection. If the exposure ends some pregnancies before delivery, outcomes are observed only for the fetuses that survive, so the estimated effects understate the true ones (Almond and Currie 2011; Rocha and Soares 2015). In other words, had those pregnancies reached term, the measured birth outcomes would have been worse. In this section I use hospital discharge records from the Mexican Ministry of Health (DGIS), as described in subsection 4.3. The data covers the network of public hospitals operated by the Ministry, which served around 60% of the population in 2010.

I use a similar specification to 2. Specifically, for municipality j in month t I estimate

$$H^k_{jt} = \beta_D \,\text{Deficit}_{jt} + \beta_S \,\text{Surplus}_{jt} + f(\text{Temp}_{jt}) + \alpha_{jq} + \delta_t + \varepsilon_{jt}, \tag{7}$$

where $H^k_{jt} = 1000 \times N^k_{jt}/B_{jt}$ is the hospitalization rate for diagnostic group k, with N^k_{jt} the discharges recorded in that group and B_{jt} the live births registered in vital statistics for the same municipality and month. I estimate three diagnostic groups: all complications, threatened preterm labor (ICD-10 O42 and O47), and abortive outcomes (O00–O08). All complications is the sum of the other two. α_{jq} are municipality $\times$ calendar-quarter effects and δ_t are year effects. Regressions are weighted by annual municipal births. Standard errors are clustered at the municipality level. The panel covers 2010–2012, since the DGIS discharge records begin in 2010.

Table 13 reports the coefficient on the rainfall deficit. Results suggest that for the whole sample the deficit induces hospitalizations due to maternal complications in general, particularly those that end in abortive outcomes and miscarriage (codes O00–O08 of the ICD-10), and those recorded as threatened preterm labor (O42, O47), that is contractions before 37 weeks without the cervical change that defines labor itself.[30] Breaking down by climate

[30]Threatened preterm labor is a warning sign rather than delivery in progress. It is coded here together with premature rupture of membranes, which is the other admission that signals the same risk and that

region reveals that the effect is concentrated in dry municipalities, where any complication rises by 5.9 admissions per 1,000 live births against a mean of 85. In non-dry municipalities the coefficients are economically and statistically insignificant.

Table 14 further disaggregates the climate regions by rural and urban subregions. Both dry subregions show significant effects on the three outcomes. The magnitudes are roughly twice as large in dry rural municipalities, where any complication rises by 9.6 per 1,000 against 5.3 in dry urban ones, and abortive outcomes by 6.4 against 3.4. These results help explain the null results found in Table 4 for this subregion, and further support the hypothesis that births are selected in dry areas during this window, and therefore the results presented in the main section can be underestimated for these regions.

supports the preterm results in the main analysis.

Table 13: Rainfall Deficits and Maternal Hospitalizations, by Climate Region

	All	Dry	Non-dry
	(1)	(2)	(3)
Panel A. Any complication			
ΔRainDef	3.867***	5.915***	0.235
	(1.073)	(1.343)	(1.596)
Mean dep. var.	86.77	85.42	87.44
Municipalities	2,431	653	1,778
Observations	82,269	21,756	60,513
Panel B. Abortive outcomes			
ΔRainDef	2.369***	3.841***	-0.731
	(0.777)	(1.005)	(1.180)
Mean dep. var.	55.80	56.61	55.40
Municipalities	2,431	653	1,778
Observations	82,269	21,756	60,513
Panel C. Threatened preterm labor			
ΔRainDef	1.551**	2.097**	1.078
	(0.636)	(0.851)	(0.885)
Mean dep. var.	30.90	28.75	31.96
Municipalities	2,431	653	1,778
Observations	82,269	21,756	60,513

Notes: Rates per 1,000 live births, winsorized at p1/p99. Municipality × calendar quarter and year fixed effects. Weighted by annual municipal births. Standard errors clustered at the municipality in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. ΔRainDef is the rainfall deficit arm, in log points. Any complication = O00–O08, O42, O47. Abortive = O00–O08. Threatened preterm labor = O42, O47. Dry = arid and semi-arid (CONAGUA).

Table 14: Rainfall Deficits and Maternal Hospitalizations, by Climate Region and Urbanization

	Dry		Non-dry	
	Rural	Urban	Rural	Urban
	(1)	(2)	(3)	(4)
Panel A. Any complication				
ΔRainDef	9.584***	5.305***	2.003	-0.041
	(2.984)	(1.468)	(3.252)	(1.795)
Mean dep. var.	91.56	84.61	74.14	92.20
Municipalities	340	313	1,035	743
Observations	10,528	11,228	33,864	26,649
Panel B. Abortive outcomes				
ΔRainDef	6.404***	3.448***	1.262	-0.946
	(2.183)	(1.092)	(2.424)	(1.325)
Mean dep. var.	59.34	56.24	45.98	58.78
Municipalities	340	313	1,035	743
Observations	10,528	11,228	33,864	26,649
Panel C. Threatened preterm labor				
ΔRainDef	3.343**	1.865**	0.755	1.049
	(1.599)	(0.950)	(1.996)	(0.997)
Mean dep. var.	31.84	28.34	27.98	33.38
Municipalities	340	313	1,035	743
Observations	10,528	11,228	33,864	26,649

Notes: Rates per 1,000 live births, winsorized at p1/p99. Municipality × calendar quarter and year fixed effects. Weighted by annual municipal births. Standard errors clustered at the municipality in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. ΔRainDef is the rainfall deficit arm, in log points. Any complication = O00–O08, O42, O47. Dry = arid and semi-arid (CONAGUA). Rural = over 50% rural population (Census 2010).

A concern with the discharge data is that in several municipalities the distribution of discharges is massed at zero. In fact, 33 percent of municipality-months record no obstetric complication of any kind, 52 percent no threatened preterm labor and 41 percent no abortive outcome. These zeros are a function of municipality size. In the smallest decile, averaging 22 births a year, 85 percent of municipality-months record no complication, against fewer than 1 percent in the largest.

To address these concerns, I estimate a Poisson model on the discharge counts, which accommodates the large number of zeros and does not require weighting.[31] Table 15 reports the coefficients, which are interpreted as semi-elasticities. Converted into admissions per 1,000 live births, they are close to the linear estimates.[32] In dry municipalities the Poisson estimates imply 5.99 additional admissions for any complication, 4.01 for abortive outcomes and 2.10 for threatened preterm labor, against 5.92, 3.84 and 2.10 in Table 13. In the full sample they imply 3.61, 2.25 and 1.41, against 3.87, 2.37 and 1.55. The results therefore do not depend on the linear form or on the weighting.

[31]The conditional mean is $\mathbb{E}[N^k_{jt} \mid \cdot] = B_{jt} \exp\big(\beta_D \text{Deficit}_{jt} + \beta_S \text{Surplus}_{jt} + f(\text{Temp}_{jt}) + \alpha_{jq} + \delta_t\big)$, with the log of live births imposed at a unit coefficient, so that β_D is the proportional response of the complication rate. I estimate it by pseudo-maximum likelihood with `ppmlhdfe` in Stata 18, clustering at the municipality level.

[32]Multiply the proportional response $e^{\beta_D} - 1$ by the baseline rate of the corresponding sample, $1000 \sum_{jt} N^k_{jt} / \sum_{jt} B_{jt}$. For any complication in dry municipalities, $(e^{0.0684} - 1) \times 84.55 = 5.99$.

Table 15: Rainfall Deficits and Maternal Hospitalizations, Poisson Pseudo-Maximum Likelihood

	All	Dry	Non-dry
	(1)	(2)	(3)
Panel A. Any complication			
ΔRainDef	0.0411***	0.0684***	0.0005
	(0.0122)	(0.0154)	(0.0166)
Implied % change	4.20	7.08	0.05
Implied per 1,000	3.61	5.99	0.04
Baseline rate	85.99	84.55	86.70
Municipalities	2,310	619	1,691
Observations	75,471	19,577	55,894
Panel B. Abortive outcomes			
ΔRainDef	0.0400***	0.0692***	-0.0130
	(0.0133)	(0.0171)	(0.0192)
Implied % change	4.08	7.17	-1.29
Implied per 1,000	2.25	4.01	-0.71
Baseline rate	55.20	55.98	54.81
Municipalities	2,266	605	1,661
Observations	72,583	18,617	53,966
Panel C. Threatened preterm labor			
ΔRainDef	0.0439**	0.0700**	0.0251
	(0.0202)	(0.0277)	(0.0263)
Implied % change	4.49	7.25	2.54
Implied per 1,000	1.41	2.10	0.82
Baseline rate	31.33	29.02	32.48
Municipalities	2,205	581	1,624
Observations	68,867	17,402	51,465

Notes: Poisson pseudo-maximum likelihood on discharge counts, with the log of live births as an offset. Unweighted. Municipality × calendar quarter and year fixed effects. Standard errors clustered at the municipality in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. ΔRainDef is the rainfall deficit arm, in log points. Its coefficient is a semi-elasticity. Italicised rows are derived from it: $100(e^{\beta} - 1)$ and $(e^{\beta} - 1)$ times the baseline rate. Baseline is the aggregate rate on the estimation sample, $1000 \sum N / \sum B$. Any complication = O00–O08, O42, O47. Abortive = O00–O08. Threatened preterm labor = O42, O47. Dry = arid and semi-arid (CONAGUA).

9 Concluding Remarks

This paper shows that rainfall deficits during gestation reduced birth weight and raised the incidence of low birth weight and preterm delivery in Mexico during the 2009–2012 drought. Identification comes from the deviation of realized rainfall from what each municipality could have received, so the comparison does not rest on differences in exposure across places. The estimates for dry municipalities should be read as lower bounds. In those areas no effect appears at delivery while maternal admissions rise, which is the pattern fetal selection produces. Outcomes are observed only for pregnancies that reach birth, so a design built on live births will miss what happened to the pregnancies that ended.

Neither agricultural disruption nor air quality appears to transmit the effect under the assumptions the mediation design requires. What the bounding exercise delivers instead is a per-unit magnitude for each channel, which can be set against the effects reported elsewhere. For particulates that magnitude is high against most published estimates, though not outside them. For crop loss no comparable benchmark exists, so the exercise ranks the two channels without settling either.

Mexico had no preventive drought plan during the episode, with the federal response consisting of emergency relief delivered once the crisis was visible. Preventive management was introduced only in 2013, after the drought had receded. Targeting a plan of that kind would have required knowing where the burden of the episode fell and what produced it, which is what this paper provides at the municipal level. Because hazard and vulnerability do not always coincide, different regions would require different interventions. Prenatal and obstetric care may be the relevant margin in dry regions, where the damage appears in the health system before the birth record. Air quality may be the relevant one in urban municipalities, where preterm delivery responds strongly.

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Appendix

A Borusyak & Hull recentering

Departing from Rocha and Soares (2015), the rainfall deficit is a measure of the form $f_{jt}(g; w)$, where the shock g is precipitation in the nine-month window preceding birth and w collects the predetermined features of the municipality's geography and climate. Writing P_{jt} for the realized nine-month average and $\bar{r}^{LT}_{j,\mathcal{M}(t)}$ for the arithmetic long-term mean over the same nine calendar months, the candidate measure is

$$f_{jt}(g; w) \;=\; \ln \bar{r}^{LT}_{j,\mathcal{M}(t)} \;-\; \ln P_{jt}. \tag{8}$$

Borusyak and Hull (2023) show that a measure of this form is generally not orthogonal to the error term even when g is exogenous, because units differ in their exposure to the shock. The bias is governed by the expected measure $\mu_{jt} = \mathbb{E}[f_{jt}(g; w) \mid w]$, and is removed by subtracting it. Their procedure is to draw counterfactual shocks from an assignment process $G(g \mid w)$, recompute the measure at each draw, and average.

In this setting the draws need not be simulated as we observe historical data for the same nine calendar months, and these historical realizations serve directly as the counterfactual shocks. Averaging (8) over the baseline years $b \in B$ gives a closed form,

$$\mu_{jt} \;=\; \ln \bar{r}^{LT}_{j,\mathcal{M}(t)} \;-\; \frac{1}{|B|} \sum_{b \in B} \ln P_{jb,\mathcal{M}(t)} \;=\; \ln \bar{r}^{LT}_{j,\mathcal{M}(t)} \;-\; \ell^{LT}_{j,\mathcal{M}(t)}, \tag{9}$$

and the recentered measure is

$$\tilde{z}_{jt} \;=\; f_{jt}(g; w) - \mu_{jt} \;=\; \ell^{LT}_{j,\mathcal{M}(t)} - \ln P_{jt} \;=\; \Delta\text{RainDef}_{jt}, \tag{10}$$

which is equation (1) in the text.

The recentered measure has two useful properties.

1. **No parallel-trends restriction is required.** Identification comes from the assignment of the rainfall shock itself, so unobserved determinants of birth outcomes are allowed to differ arbitrarily across municipalities. Households may sort into locations on climate, geography, agricultural structure, or any other predetermined characteristic, observed or not, and identification is unaffected.

2. **The counterfactuals are observed.** Borusyak and Hull (2023) obtain μ_{jt} by simulating counterfactual shocks from a specified assignment process. Here the same nine calendar months recur in every baseline year, so the historical realizations serve as the counterfactuals and the expectation has the closed form in (9).

B Tables

B.1 Descriptives

Table B.1: Residual Variation in the Rainfall Deficit Measure

Rainfall deficit, Deficit (N = 8,102,177)
Mean (SD) = 0.176 (0.249)

Fixed effects	R^2	σ_e	*Share of observations (%) where $\lvert\hat{e}\rvert$ exceeds threshold*				
			0.10 (10%)	0.15 (14%)	0.25 (22%)	0.50 (39%)	1.00 (63%)
Surplus arm only, no FE	0.142	0.231	61.8	43.0	11.5	4.1	0.6
Municipality-level geography							
Muni × quarter, year	0.467	0.182	48.0	30.8	12.0	1.6	0.2
Muni × quarter, year, state trend†	0.516	0.173	46.4	28.3	10.4	1.4	0.2
Muni × month, year	0.521	0.172	46.2	29.2	11.3	1.6	0.1
Muni × month, year, state trend	0.570	0.163	44.7	26.8	9.6	1.3	0.1
State-level geography							
State × quarter, year	0.421	0.189	49.9	33.1	13.0	1.8	0.2
State × quarter, year, state trend	0.470	0.181	48.1	31.0	11.5	1.7	0.2
State × month, year	0.466	0.182	48.4	32.2	12.5	1.8	0.1
State × month, year, state trend	0.514	0.173	46.8	29.7	11.1	1.6	0.1

Notes: Each row regresses the measure on the indicated fixed effects alone, no controls, on individual births 2009–2012. Deficit = $\max(\Delta\text{Raindef}, 0)$ is the deficit arm of the 9-month BH log-deviation split at the municipality long-term norm. R^2 and σ_e are from each regression. Remaining columns give the percentage of observations with $|\hat{e}|$ above the column threshold. Thresholds are in log-deviation units; parentheses give the implied rainfall reduction, $1 - e^{-x}$. The surplus arm enters every row as a control and is not reported; the reported residual is therefore the variation identifying the deficit coefficient. † preferred specification in the main results.

Table B.2: Drought Categories and Characteristics in the MSM

Drought Category	**Percent of Occurrence**	**Characteristics and Potential Effects**
D0 - Anomalía seca (Abnormally dry)	20% to <30%	Pre-drought conditions or lingering effects post-drought. Can affect agriculture, soil moisture, and air quality.
D1 - Sequía moderada (Moderate drought)	10% to <20%	Possible crop/pasture damage, low water supplies, and voluntary water-use restrictions.
D2 - Sequía severa (Severe drought)	5% to <10%	Likely crop/pasture losses, water shortages, and some mandatory water-use restrictions.
D3 - Sequía extrema (Extreme drought)	2% to <5%	Major crop/pasture losses, widespread water shortages, and mandatory water-use restrictions.
D4 - Sequía excepcional (Exceptional drought)	<2%	Widespread crop and pasture failures, water emergencies, and severe water-use restrictions.

Table B.3: Monthly transitions between drought categories (2009–2012)

	D00	D0	D1	D2	D3	D4	Total
D00	91.5%	6.9%	1.4%	0.2%	0.1%	0.0%	100%
	64,266	4,832	964	152	50	0	70,264
D0	25.8%	58.6%	12.4%	2.8%	0.4%	0.0%	100%
	4,706	10,694	2,255	502	79	0	18,236
D1	11.8%	15.0%	59.8%	10.6%	2.8%	0.0%	100%
	1,673	2,117	8,466	1,504	394	1	14,155
D2	2.9%	4.3%	19.9%	58.0%	14.6%	0.4%	100%
	250	366	1,707	4,986	1,251	31	8,591
D3	0.3%	0.9%	7.8%	31.5%	52.8%	6.7%	100%
	11	38	332	1,333	2,234	282	4,230
D4	0.0%	1.1%	3.0%	9.5%	33.9%	52.5%	100%
	0	7	20	63	224	347	661

Notes: Transitions are computed month to month within municipality. Each percentage is conditional on the origin drought category in month t. Counts report the corresponding number of municipality-month transitions.

B.2 Mismatch SINAC and Vital Statistics

Table B.4: Missing birth certificates by state (2009)

State	Coverage rate
Aguascalientes	100.7%
Tabasco	99.4%
Tamaulipas	98.9%
Baja California	98.0%
Guanajuato	97.5%
Coahuila	95.5%
Tlaxcala	95.0%
Colima	95.0%
Quintana Roo	94.5%
México	93.3%
Querétaro	93.1%
Sinaloa	92.5%
Yucatán	92.4%
Baja California Sur	92.2%
Sonora	90.4%
Morelos	90.1%
Nayarit	89.8%
San Luis Potosí	89.4%
Hidalgo	88.4%
Durango	88.1%
Jalisco	87.3%
Ciudad de México	86.7%
Zacatecas	86.4%
Chihuahua	86.1%
Michoacán	86.1%
Campeche	85.3%
Puebla	84.1%
Nuevo León	82.3%
Veracruz	78.7%
Oaxaca	64.9%
Guerrero	64.6%
Chiapas	53.8%

B.3 Fully interacted model of rainfall deficit

Table B.5 estimates the four subgroup effects jointly, regressing each outcome on the deficit arm, its interaction with a dry-climate indicator, its interaction with a rural indicator, and the triple interaction of the three. The surplus arm enters with the same interaction set and is not reported. The group-specific estimates at the foot of the table recover the separate regressions of Table 4 for all three outcomes. Preterm is the one column where the two tables differ by more than rounding, since the joint specification holds the sample at the births with valid birth weight while the separate regressions let preterm use its own larger sample.

The formal tests say which of these differences are statistically distinguishable. For birth weight the rural damage is higher in non-dry municipalities, where rural areas lose significantly more than urban ones. The same gap within dry municipalities is insignificant. Read along the climate margin the result appears as an attenuation, with the dry zone attenuating the effect among rural municipalities and doing nothing among urban ones. The triple interaction is itself significant, so the rural penalty belongs to the non-dry zone specifically and is not a general rural effect. Low birth weight follows the same structure. For preterm delivery none of the four interactions reaches significance, so the subgroup pattern visible in the point estimates cannot be distinguished statistically and should not be pressed.

B.4 Robustness of the main results

Table B.5: Differential Effects of Rainfall Deficits by Climate Zone and Urbanization

	Weight (g) (1)	LBW (2)	Preterm (<37) (3)
Regression estimates			
Deficit: β_1	-7.491***	0.0026***	0.0041***
	(1.946)	(0.0009)	(0.0012)
Deficit × Dry: β_2	0.513	-0.0013	-0.0019
	(2.635)	(0.0011)	(0.0014)
Deficit × Rural: β_3	-13.413***	0.0036	0.0009
	(4.632)	(0.0022)	(0.0024)
Deficit × Dry × Rural: β_4	18.922***	-0.0069**	-0.0016
	(6.437)	(0.0030)	(0.0032)
Fixed effects	Muni × Quarter, Year, State × Trend		
Other controls	Temperature (mean + quadratic), Demographics		
Observations	7,470,462	7,470,462	7,470,462
Formal tests			
Dry gap, Urban:	0.5	-0.0013	-0.0019
H_0: $\beta_2 = 0$	(2.6)	(0.0011)	(0.0014)
Dry gap, Rural:	19.4***	-0.0082***	-0.0035
H_0: $\beta_2 + \beta_4 = 0$	(5.9)	(0.0028)	(0.0028)
Rural gap, Non-dry:	-13.4***	0.0036	0.0009
H_0: $\beta_3 = 0$	(4.6)	(0.0022)	(0.0024)
Rural gap, Dry:	5.5	-0.0033	-0.0007
H_0: $\beta_3 + \beta_4 = 0$	(4.5)	(0.0020)	(0.0021)
Group-specific estimates			
{Non-dry, Urban}	-7.5***	0.0026***	0.0041***
H_0: $\beta_1 = 0$	(1.9)	(0.0009)	(0.0012)
{Dry, Urban}	-7.0***	0.0013**	0.0022***
H_0: $\beta_1 + \beta_2 = 0$	(2.0)	(0.0007)	(0.0008)
{Non-dry, Rural}	-20.9***	0.0062***	0.0050**
H_0: $\beta_1 + \beta_3 = 0$	(4.3)	(0.0020)	(0.0021)
{Dry, Rural}	-1.5	-0.0020	0.0015
H_0: $\beta_1 + \beta_2 + \beta_3 + \beta_4 = 0$	(4.1)	(0.0019)	(0.0020)

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. All columns report estimates from a single regression of the outcome on the rainfall deviation index split at the municipality long-term norm, its interactions with Dry (Koppen arid/semi-arid) and Rural (>50% rural population), and the triple interaction. Surplus carries the same interaction set and is estimated but not reported. Municipality × quarter, year, and state-specific linear trend fixed effects included. Individual controls: maternal age, age^2, marital status, parity, schooling. Mean-centered 9-month temperature and its square, interacted with the climate × urbanization group. Standard errors clustered at municipality × quarter in parentheses. report linear combinations testing whether the drought effect differs across groups. report the implied total drought effect for each municipality type.

Table B.6: Rainfall Deficits and Birth Outcomes. Excl. Underreporting States

	Dependent Variable		
	Weight (g)	LBW	Preterm (< 37)
	(1)	(2)	(3)
Deficit	-7.242***	0.002***	0.003***
	(1.435)	(0.001)	(0.001)
Surplus	-4.535	0.002*	0.002
	(2.842)	(0.001)	(0.001)
Observations	6,396,764	6,396,764	6,396,764
Mean dep. var.	3159.278	0.058	0.063

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Excludes Chiapas, Guerrero, Oaxaca and Veracruz (2009 SINAC coverage below 80 percent of INEGI vital statistics). Municipality × calendar-quarter FE, year FE, state-specific linear trends. Controls: mother's age, age^2, marital status, parity, schooling; mean-centered 9-month temperature and its square. SE clustered at municipality×quarter. Deficit and Surplus are the two sides of the 9-month log-deviation, split at the municipality long-term norm, both non-negative.

Table B.7: Lagged, Contemporaneous, and Lead Rainfall Deviations

	Weight (g)			LBW			Preterm (< 37)		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
Deficit_{t-1}	-5.0547***	-2.7719**		0.0019***	0.0013**		0.0020***	0.0010	
	(1.2289)	(1.2266)		(0.0006)	(0.0005)		(0.0007)	(0.0007)	
Surplus_{t-1}	-4.9335*	-5.2291**		0.0004	0.0007		0.0007	0.0014	
	(2.7685)	(2.4727)		(0.0011)	(0.0010)		(0.0013)	(0.0012)	
Deficit_{t}	-8.7654***			0.0024***			0.0035***		
	(1.3954)			(0.0006)			(0.0007)		
Surplus_{t}	-5.1595*			0.0013			0.0009		
	(2.9315)			(0.0012)			(0.0014)		
Deficit_{t+1}	0.0592		1.9396	0.0001		-0.0004	-0.0002		-0.0009
	(1.4350)		(1.3900)	(0.0006)		(0.0005)	(0.0007)		(0.0006)
Surplus_{t+1}	-2.4725		-1.5515	-0.0006		-0.0006	-0.0006		-0.0004
	(2.5805)		(2.3635)	(0.0011)		(0.0010)	(0.0013)		(0.0012)
Observations	7,470,462	7,470,462	7,470,462	7,470,462	7,470,462	7,470,462	7,470,462	7,470,462	7,470,462
Mean of Y (2009)	3170.2680	3170.2680	3170.2680	0.0546	0.0546	0.0546	0.0604	0.0604	0.0604

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Municipality × calendar-quarter FE, year FE, state-specific linear trends. Demographic controls and a quadratic in mean-centered 9-month temperature. SE clustered at municipality-quarter. Deficit and Surplus are the two sides of the log-deviation, split at the municipality long-term norm, both non-negative, each temporal term split separately. Lag $(t-1)$ and lead $(t+1)$ are the same calendar month in the previous and next year. Columns (1) lag, contemporaneous and lead jointly; (2) lag only; (3) lead only. Common sample anchored on column (1) for birth weight. Means for 2009.

Table B.8: Rainfall Deviations and Birth Outcomes: Alternative Exposure Windows

	Weight (g)			LBW			Preterm (< 37)		
	3m	6m	9m	3m	6m	9m	3m	6m	9m
Deficit	-0.1021	-1.9937***	-7.7754***	0.0002	0.0009***	0.0021***	0.0002	0.0010***	0.0036***
	(0.3294)	(0.5669)	(1.5878)	(0.0002)	(0.0003)	(0.0006)	(0.0002)	(0.0003)	(0.0007)
1-SD effect	[-0.1068]	[-1.1680]	[-2.3552]	[0.0002]	[0.0006]	[0.0006]	[0.0002]	[0.0006]	[0.0011]
Surplus	-0.5693	-3.6389**	-3.5003	-0.0002	0.0017**	0.0016	-0.0006*	0.0012	0.0016
	(0.6119)	(1.7247)	(2.6759)	(0.0003)	(0.0008)	(0.0011)	(0.0003)	(0.0008)	(0.0013)
1-SD effect	[-0.5954]	[-2.1318]	[-1.0603]	[-0.0002]	[0.0010]	[0.0005]	[-0.0006]	[0.0007]	[0.0005]
Observations	7,470,462	7,470,462	7,470,462	7,470,462	7,470,462	7,470,462	7,470,462	7,470,462	7,470,462
Mean of Y (2009)	3170.2680	3170.2680	3170.2680	0.0546	0.0546	0.0546	0.0604	0.0604	0.0604
SD of deviation	1.0458	0.5858	0.3029	1.0458	0.5858	0.3029	1.0458	0.5858	0.3029
Residual SD, deviation	0.9120	0.4618	0.2150	0.9120	0.4618	0.2150	0.9120	0.4618	0.2150
Residual SD, deficit arm	0.6986	0.3958	0.1667	0.6986	0.3958	0.1667	0.6986	0.3958	0.1667
Residual SD, surplus arm	0.3546	0.1504	0.0909	0.3546	0.1504	0.0909	0.3546	0.1504	0.0909

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Log-deviation rainfall measure over the indicated window (3, 6 or 9 months prior to birth), winsorized at p1/p99 within window, then split at the municipality long-term norm. Deficit and Surplus are the two sides of the log-deviation, both non-negative. Bracketed 1-SD effects multiply each coefficient by the within-sample SD of the underlying deviation. Municipality × quarter FE, year FE, state-specific linear trends. Demographic controls and a quadratic in mean-centered temperature over the matching window. SE clustered at municipality-quarter. Common sample anchored on the 3-month specification for birth weight. Means for 2009. Assembled by hand from 14rob_bh_windows_12col, dropping the 12-month window.

B.5 Agricultural Mediation

B.6 Air Pollution Mediation

C Figures

C.1 Robustness of the main results

Table B.9: Rainfall Deficits and Birth Outcomes (Conley SEs)

	Weight (g)		LBW		Preterm (< 37)	
	150 km (1)	200 km (2)	150 km (3)	200 km (4)	150 km (5)	200 km (6)
Panel A. Rainfall deficit						
Deficit	-7.502***	-7.502***	0.002***	0.002***	0.003***	0.003***
	(1.467)	(1.575)	(0.001)	(0.001)	(0.001)	(0.001)
Surplus	-3.942	-3.942	0.002	0.002	0.001	0.001
	(2.666)	(2.767)	(0.001)	(0.001)	(0.001)	(0.001)
Births (sum of weights)	8,101,555	8,101,555	8,101,555	8,101,555	8,101,555	8,101,555
Mean dep. var.	3162.665	3162.665	0.056	0.056	0.067	0.067
Panel B. Drought categories						
D0	-2.441***	-2.441***	0.001***	0.001***	0.001**	0.001**
	(0.834)	(0.826)	(0.000)	(0.000)	(0.000)	(0.000)
D1	-3.200***	-3.200***	0.001**	0.001**	0.001**	0.001**
	(0.942)	(0.958)	(0.000)	(0.000)	(0.000)	(0.001)
D2	-4.011***	-4.011***	0.001**	0.001**	0.001	0.001
	(1.143)	(1.216)	(0.000)	(0.000)	(0.001)	(0.001)
D3–D4	-5.747***	-5.747***	0.001**	0.001**	0.001*	0.001
	(1.529)	(1.613)	(0.001)	(0.001)	(0.001)	(0.001)
Births (sum of weights)	8,101,555	8,101,555	8,101,555	8,101,555	8,101,555	8,101,555
Mean dep. var.	3162.665	3162.665	0.056	0.056	0.067	0.067

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Municipality × birth-year-month cells, weighted by births in the cell. Municipality × calendar-quarter FE, year FE, state-specific linear trends. Cell-mean controls: mother's age, age^2, share married, mean parity, mean schooling; mean-centered 9-month temperature and its square. SE corrected for spatial dependence within the indicated cutoff (Conley 1999), via acreg (Colella et al. 2019). Point estimates are identical across cutoffs; only SEs differ. Deficit and Surplus are the two sides of the 9-month log-deviation, split at the municipality long-term norm, both non-negative. Panel B: drought-category indicators; omitted = No Drought.

Table B.10: Agricultural Mediation, Individual Births: Birth Weight

	OLS			IV	
				First stage	Second stage
Dep. variable	BW	BW	BW	CL	BW
Deficit	-10.072***	-9.699***	-9.661***	8.458***	-10.940***
	(1.538)	(1.507)	(1.503)	(1.232)	(1.916)
Deficit × Pre-Rainfed		0.046	0.048	0.315***	
		(0.046)	(0.047)	(0.052)	
Crop loss (%)			-0.005		0.147
			(0.018)		(0.135)
KP F-statistic					37.07
AR p ($\beta = 0$)					0.311
Direct effect					-10.940
Indirect effect					1.241
Total effect					-9.699
Observations	6,665,565	6,665,565	6,665,565	6,665,565	6,665,565

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012. CL: crop loss rate in percent, SIAP, municipality-year. Deficit: size of the 9-month BH rainfall deficit. The surplus arm enters every column as a control, not reported; its Pre-Rainfed interaction is excluded from this specification. Pre-Rainfed: 2005–2008 rainfed share of area sown, centred at 72.72 percent. All columns interact both arms with rural share, log population, dry indicator, and a deprivation index. Column (5) instruments CL with Deficit × Pre-Rainfed. Indirect effect: CL coefficient × first-stage Deficit; total = direct + indirect. AR: Anderson-Rubin test of $\beta = 0$. FE: municipality × quarter, year, state trends. Controls: maternal age and its square, marital status, parity, schooling, temperature quadratic. SE clustered by municipality.

Table B.11: Agricultural Mediation, Individual Births: Low Birth Weight

	OLS			IV	
				First stage	Second stage
Dep. variable	LBW	LBW	LBW	CL	LBW
Deficit	0.0026***	0.0026***	0.0025***	8.458***	0.0026***
	(0.0006)	(0.0007)	(0.0007)	(1.232)	(0.0007)
Deficit × Pre-Rainfed		0.0000	-0.0000	0.315***	
		(0.0000)	(0.0000)	(0.052)	
Crop loss (%)			0.00001		0.00000
			(0.00001)		(0.00005)
KP F-statistic					37.07
AR p ($\beta = 0$)					0.996
Direct effect					0.0026
Indirect effect					0.0000
Total effect					0.0026
Observations	6,665,565	6,665,565	6,665,565	6,665,565	6,665,565

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012. CL: crop loss rate in percent, SIAP, municipality-year. Deficit: size of the 9-month BH rainfall deficit. The surplus arm enters every column as a control, not reported; its Pre-Rainfed interaction is excluded from this specification. Pre-Rainfed: 2005–2008 rainfed share of area sown, centred at 72.72 percent. All columns interact both arms with rural share, log population, dry indicator, and a deprivation index. Column (5) instruments CL with Deficit × Pre-Rainfed. Indirect effect: CL coefficient × first-stage Deficit; total = direct + indirect. AR: Anderson-Rubin test of $\beta = 0$. FE: municipality × quarter, year, state trends. Controls: maternal age and its square, marital status, parity, schooling, temperature quadratic. SE clustered by municipality.

Table B.12: Agricultural Mediation, Individual Births: Preterm Birth

	OLS			IV	
				First stage	Second stage
Dep. variable	PT	PT	PT	CL	PT
Deficit	0.0036***	0.0036***	0.0036***	8.458***	0.0037***
	(0.0008)	(0.0008)	(0.0008)	(1.232)	(0.0008)
Deficit × Pre-Rainfed		-0.0000	-0.0000	0.315***	
		(0.0000)	(0.0000)	(0.052)	
Crop loss (%)			0.00001		-0.00001
			(0.00001)		(0.00006)
KP F-statistic					37.07
AR p ($\beta = 0$)					0.866
Direct effect					0.0037
Indirect effect					-0.0001
Total effect					0.0036
Observations	6,665,565	6,665,565	6,665,565	6,665,565	6,665,565

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012. CL: crop loss rate in percent, SIAP, municipality-year. Deficit: size of the 9-month BH rainfall deficit. The surplus arm enters every column as a control, not reported; its Pre-Rainfed interaction is excluded from this specification. Pre-Rainfed: 2005–2008 rainfed share of area sown, centred at 72.72 percent. All columns interact both arms with rural share, log population, dry indicator, and a deprivation index. Column (5) instruments CL with Deficit × Pre-Rainfed. Indirect effect: CL coefficient × first-stage Deficit; total = direct + indirect. AR: Anderson-Rubin test of $\beta = 0$. FE: municipality × quarter, year, state trends. Controls: maternal age and its square, marital status, parity, schooling, temperature quadratic. SE clustered by municipality.

Table B.13: Drought, Crop Loss and Birth Outcomes: Spring-Summer Share Instrument

	First stage	IV second stage		
	(1)	(2)	(3)	(4)
Dep. variable	CL	BW	LBW	PT
Deficit (direct)	7.452***	-11.270***	0.0024***	0.0035***
	(1.277)	(2.452)	(0.0007)	(0.0008)
Deficit × Pre-PV share	0.289***			
	(0.053)			
Crop loss (%)		0.202	0.00002	0.00002
		(0.229)	(0.00006)	(0.00007)
Indirect effect		1.508	0.0001	0.0001
		(1.705)	(0.0004)	(0.0005)
Total effect		-9.762***	0.0026***	0.0037***
		(1.469)	(0.0006)	(0.0008)
KP F-statistic	29.63			
AR p ($\beta = 0$)		0.380	0.727	0.808
Observations	6,665,565	6,665,565	6,665,565	6,665,565

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012. CL: crop loss rate (%), SIAP municipality-year, instrumented in (2)–(4) with Deficit × Pre-PV share. Deficit: 9-month BH rainfall deficit. Pre-PV share: 2005–08 share of sown area in the spring-summer cycle, centred at 80.29%. Surplus arm and the moderator interactions are unreported controls. IE = CL coefficient × first-stage Deficit. TE = direct + indirect. AR: Anderson-Rubin test of $\beta = 0$. FE: municipality × quarter, year, state trends. Controls: maternal age, age^2, marital status, parity, schooling, temperature quadratic. SE clustered by municipality.

Table B.14: Agricultural Mediation, Overidentified: Crop Loss Instrumented with Both Interactions

	First stage	Second stage		
	(1)	(2)	(3)	(4)
Dep. variable	CL	BW	LBW	PT
Deficit (direct)	8.661***	-11.030***	0.0025***	0.0036***
	(1.255)	(2.035)	(0.0007)	(0.0008)
Deficit × Pre-Rainfed	0.249***			
	(0.061)			
Deficit × Pre-PV	0.138**			
	(0.067)			
Crop loss (%)		0.163	0.00001	-0.00000
		(0.149)	(0.00005)	(0.00006)
Indirect effect		1.408	0.0000	-0.0000
		(1.291)	(0.0004)	(0.0005)
Total effect		-9.632***	0.0026***	0.0036***
		(1.475)	(0.0007)	(0.0008)
KP F-statistic	22.00			
Hansen J p		0.736	0.649	0.608
AR p ($\beta = 0$)		0.579	0.896	0.857
Observations	6,665,565	6,665,565	6,665,565	6,665,565

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012. CL: crop loss rate (%), SIAP municipality-year; instrumented in (2)–(4) with both interactions. Deficit: 9-month BH rainfall deficit. Pre-Rainfed: 2005–08 rainfed share of sown area, centred at 72.72%. Pre-PV: 2005–08 spring-summer share, centred at 80.29%. Surplus arm and its interactions: unreported controls. Both arms interacted with rural share, log population, dry indicator, deprivation index. IE = CL coefficient × first-stage Deficit, SE holding the first stage fixed. TE and its SE from the reduced form, where the Deficit coefficient equals direct + indirect. KP F common to (2)–(4). AR: Anderson-Rubin, $\beta = 0$. J: Hansen, 1 df. FE: municipality × quarter, year, state trends. Controls: maternal age, age^2, marital status, parity, schooling, temperature quadratic. SE clustered by municipality.

Table B.15: Drought, Production Volume and Birth Outcomes

	First stage	IV second stage		
	(1)	(2)	(3)	(4)
Dep. variable	Volume	BW	LBW	PT
Deficit (direct)	-0.142***	-11.504***	0.0026***	0.0037***
	(0.040)	(2.417)	(0.0008)	(0.0009)
Deficit × Pre-Rainfed	-0.0036***			
	(0.0009)			
Log volume		-12.674	-0.00002	0.00080
		(12.967)	(0.00411)	(0.00482)
Indirect effect		1.805	0.0000	-0.0001
		(1.847)	(0.0006)	(0.0007)
Total effect		-9.699***	0.0026***	0.0036***
		(1.581)	(0.0007)	(0.0008)
KP F-statistic	15.78			
AR p ($\beta = 0$)		0.311	0.996	0.866
Observations	6,665,565	6,665,565	6,665,565	6,665,565

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012. Volume: log production volume, annual crops, both cycles, SIAP municipality-year, instrumented in (2)–(4) with Deficit × Pre-Rainfed. Deficit: 9-month BH rainfall deficit. Pre-Rainfed: 2005–08 rainfed share of sown area, centred at 72.72%. Surplus arm and the moderator interactions are unreported controls. IE = volume coefficient × first-stage Deficit. TE = direct + indirect. AR: Anderson-Rubin test of $\beta = 0$. FE: municipality × quarter, year, state trends. Controls: maternal age, age^2, marital status, parity, schooling, temperature quadratic. SE clustered by municipality.

Table B.16: Pollution Mediation, Individual Births: Birth Weight

	OLS			IV	
				First stage	Second stage
Dep. variable	BW	BW	BW	$PM_{2.5}$	BW
Deficit	-8.921***	-8.797***	-9.001***	0.734***	-9.364***
	(1.391)	(1.367)	(1.362)	(0.147)	(2.605)
Deficit × Log cars		0.909	0.582	1.178***	
		(3.578)	(3.573)	(0.236)	
$PM_{2.5}$			0.278***		0.772
			(0.092)		(3.030)
KP F-statistic					24.97
AR p ($\beta = 0$)					0.799
Direct effect					-9.364
Indirect effect					0.567
Total effect					-8.797
Observations	7,262,572	7,262,572	7,262,572	7,262,572	7,262,572

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012. $PM_{2.5}$: municipal mean in $\mu g/m^3$, month of birth, instrumented in column (5) with Deficit × Log cars. Deficit: 9-month BH rainfall deficit. Log cars: 2009 registered cars per 1,000 residents, in logs, centred at 4.896. Fleet controls in columns (2)–(5): the bus, freight, motorcycle and public-service rates, on the same scale. Both arms interacted with rural share, log population, dry indicator and a census deprivation index; surplus arm and these interactions unreported. IE = $PM_{2.5}$ coefficient × first-stage Deficit. TE = direct + indirect. KP F: Kleibergen-Paap rk Wald. AR: Anderson-Rubin test of $\beta = 0$. FE: municipality × quarter, year, state trends. Controls: maternal age, age^2, marital status, parity, schooling, temperature quadratic. SE clustered by municipality × quarter.

Table B.17: Pollution Mediation, Individual Births: Low Birth Weight

	OLS			IV	
Dep. variable	LBW	LBW	LBW	First stage $PM_{2.5}$	Second stage LBW
Deficit	0.0024***	0.0025***	0.0025***	0.734***	0.0029***
	(0.0006)	(0.0006)	(0.0006)	(0.147)	(0.0010)
Deficit × Log cars		-0.0006	-0.0006	1.178***	
		(0.0013)	(0.0013)	(0.236)	
$PM_{2.5}$			-0.00003		-0.00054
			(0.00005)		(0.00112)
KP F-statistic					24.97
AR p ($\beta = 0$)					0.632
Direct effect					0.0029
Indirect effect					-0.0004
Total effect					0.0025
Observations	7,262,572	7,262,572	7,262,572	7,262,572	7,262,572

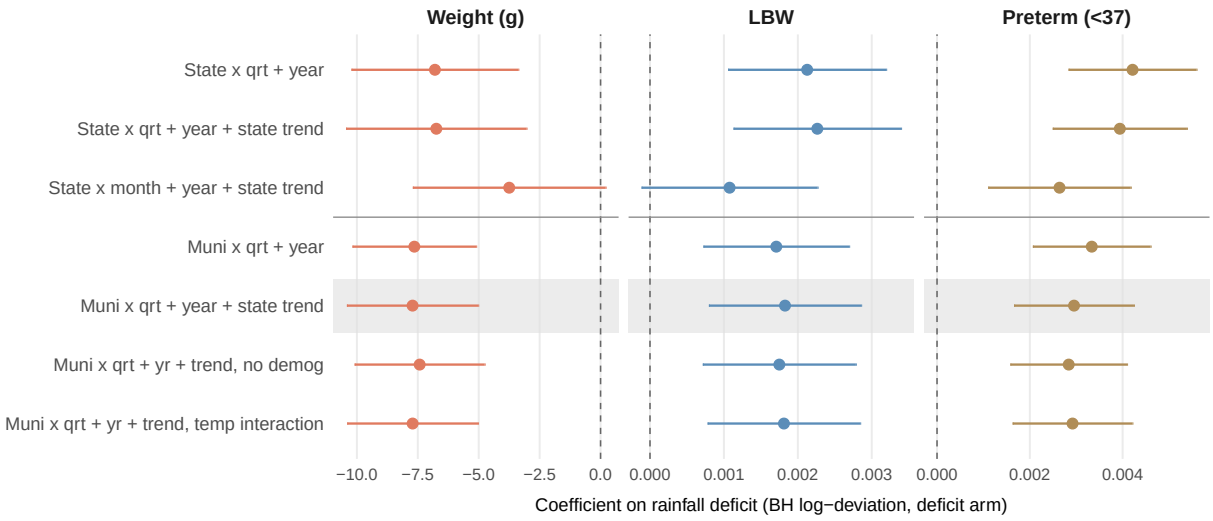
Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012. $PM_{2.5}$: municipal mean in $\mu g/m^3$, month of birth, instrumented in column (5) with Deficit × Log cars. Deficit: 9-month BH rainfall deficit. Log cars: 2009 registered cars per 1,000 residents, in logs, centred at 4.896. Fleet controls in columns (2)–(5): the bus, freight, motorcycle and public-service rates, on the same scale. Both arms interacted with rural share, log population, dry indicator and a census deprivation index; surplus arm and these interactions unreported. IE = $PM_{2.5}$ coefficient × first-stage Deficit. TE = direct + indirect. KP F: Kleibergen-Paap rk Wald. AR: Anderson-Rubin test of $\beta = 0$. FE: municipality × quarter, year, state trends. Controls: maternal age, age^2, marital status, parity, schooling, temperature quadratic. SE clustered by municipality × quarter.

Table B.18: Pollution Mediation, Individual Births: Preterm Birth

	OLS			IV	
Dep. variable	PT	PT	PT	First stage $PM_{2.5}$	Second stage PT
Deficit	0.0034***	0.0037***	0.0036***	0.734***	0.0037***
	(0.0008)	(0.0008)	(0.0008)	(0.147)	(0.0013)
Deficit × Log cars		-0.0001	-0.0001	1.178***	
		(0.0016)	(0.0016)	(0.236)	
$PM_{2.5}$			0.00001		-0.00010
			(0.00005)		(0.00136)
KP F-statistic					24.97
AR p ($\beta = 0$)					0.941
Direct effect					0.0037
Indirect effect					-0.0001
Total effect					0.0037
Observations	7,262,572	7,262,572	7,262,572	7,262,572	7,262,572

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Individual births, SINAC 2009–2012. $PM_{2.5}$: municipal mean in $\mu g/m^3$, month of birth, instrumented in column (5) with Deficit × Log cars. Deficit: 9-month BH rainfall deficit. Log cars: 2009 registered cars per 1,000 residents, in logs, centred at 4.896. Fleet controls in columns (2)–(5): the bus, freight, motorcycle and public-service rates, on the same scale. Both arms interacted with rural share, log population, dry indicator and a census deprivation index; surplus arm and these interactions unreported. IE = $PM_{2.5}$ coefficient × first-stage Deficit. TE = direct + indirect. KP F: Kleibergen-Paap rk Wald. AR: Anderson-Rubin test of $\beta = 0$. FE: municipality × quarter, year, state trends. Controls: maternal age, age^2, marital status, parity, schooling, temperature quadratic. SE clustered by municipality × quarter.

Figure C.1: Rainfall Deficit and Birth Outcomes across Fixed-Effect Structures



Notes: Each point is the coefficient on Deficit, the size of the rainfall deficit, from a separate regression. Horizontal bars are 95 percent confidence intervals. Rows vary the fixed-effect structure, the demographic controls and the temperature interaction. The shaded row is the preferred specification reported in Table 2. Standard errors clustered at municipality × quarter throughout, including the state-level rows. Individual births, SINAC 2009–2012.